PROGRAMA DE PÓS-GRADUAÇÃO EM

Biologia Vegetal

Universidade Federal de Pernambuco

Modelos de polímeros

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Leibniz Institute of Plant Genetics and Crop Plant Research



26 de outubro de 2026

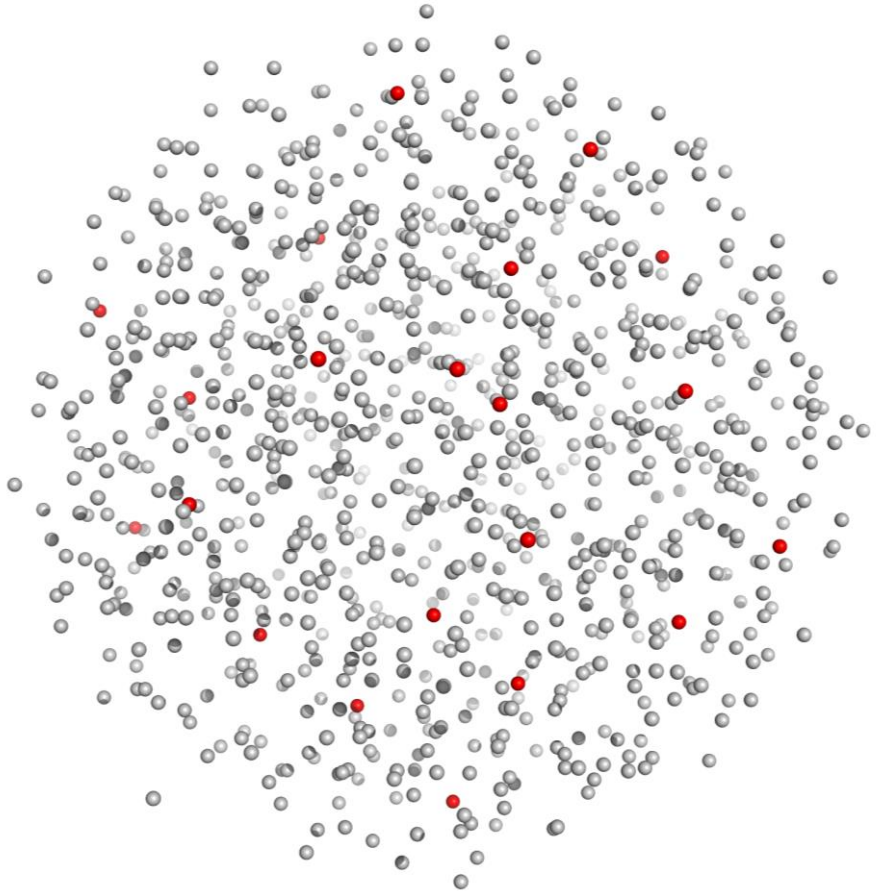
In Silico Genebank Proteomics

IPK LEIBNIZ-INSTITUT

Um experimento mental

E se o genoma fosse feito de muitas moléculas curtas de DNA?

Associação por difusão



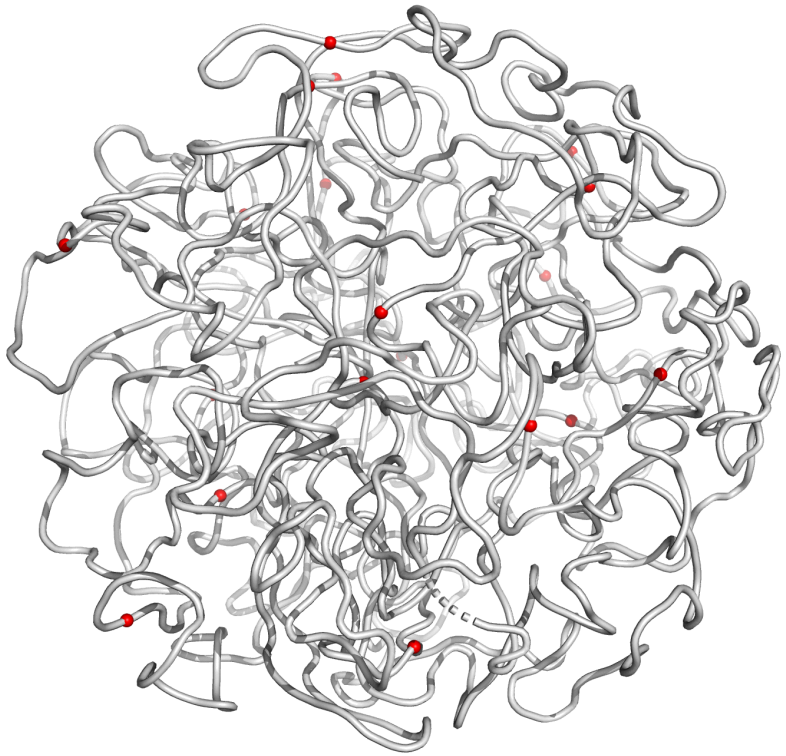
Proteínas teriam que contar com a sorte para encontrar seu sítio de DNA e seus reguladores.

Como as proteínas fazem:
Taxa de associação ~ 10^6-10^7 /Ms

Repressor Lac e seu sítio de DNA:
Taxa de associação ~ 10^{10} /Ms

Riggs et al., 1970

Difusão de polímero

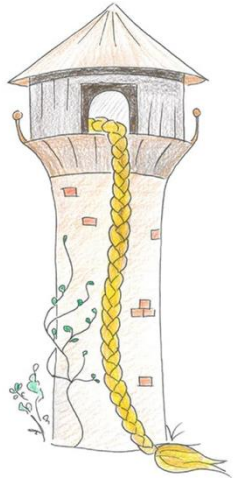


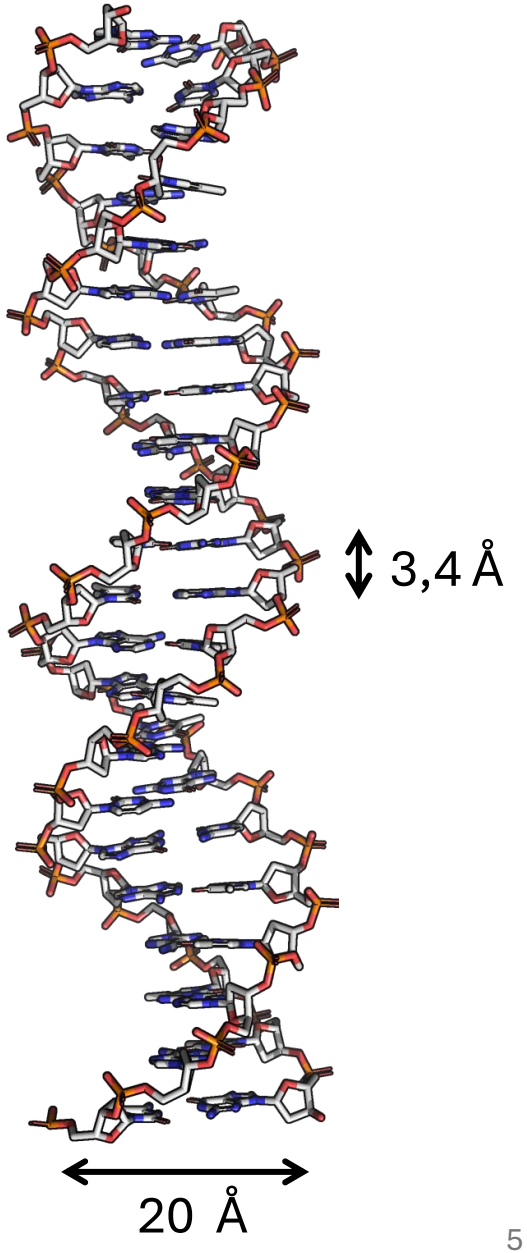
Difusão de polímero de DNA é muito mais lenta:
Taxa de associação ~ 10^8 /Ms

Mirny et al., 2009

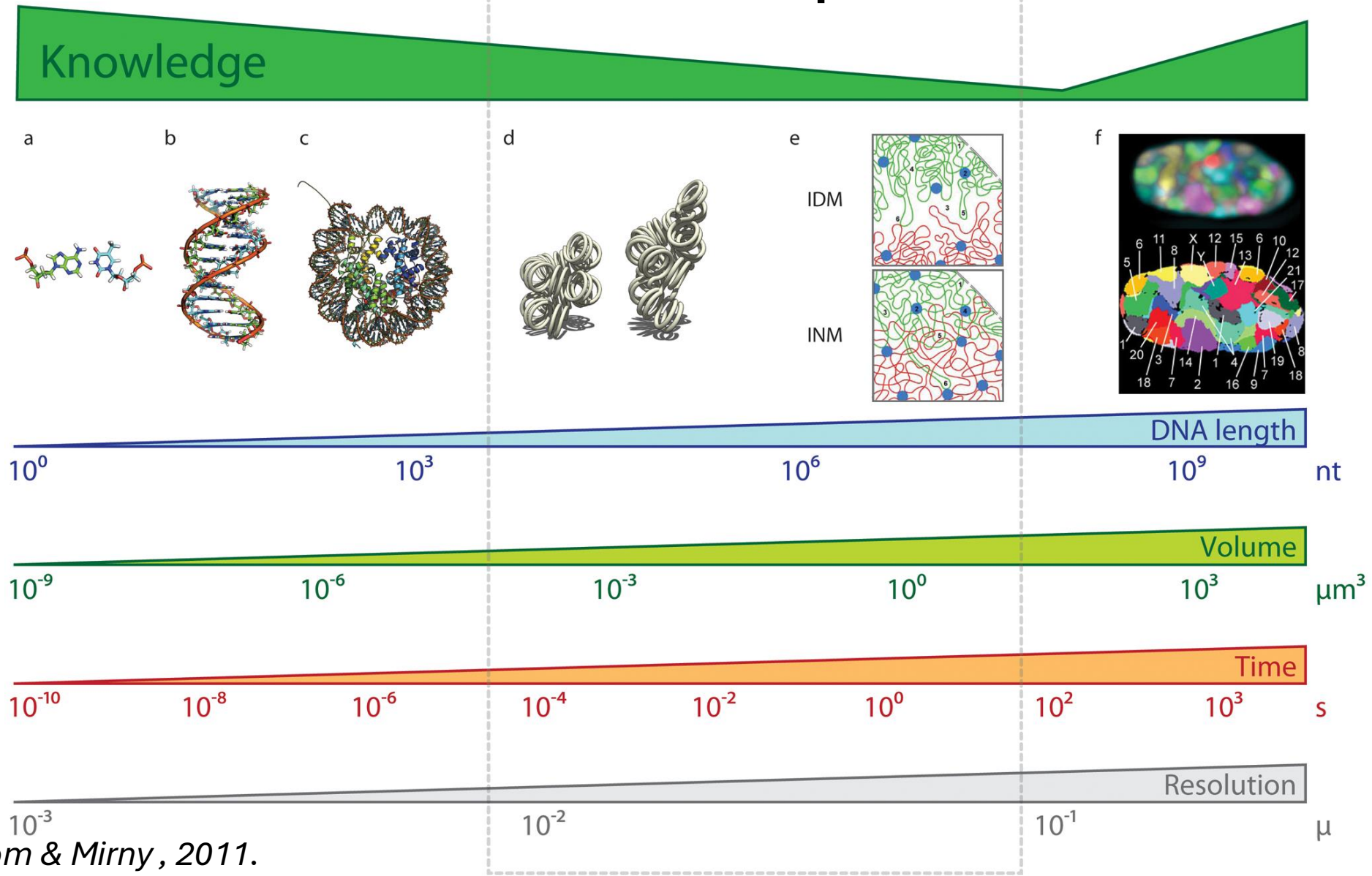
Um polímero muito muito longo

Objeto	Espessura	Comprimento
1 Mbp DNA	20 Å	0,34 mm
Um fio de cabelo	0,1 mm	17 m



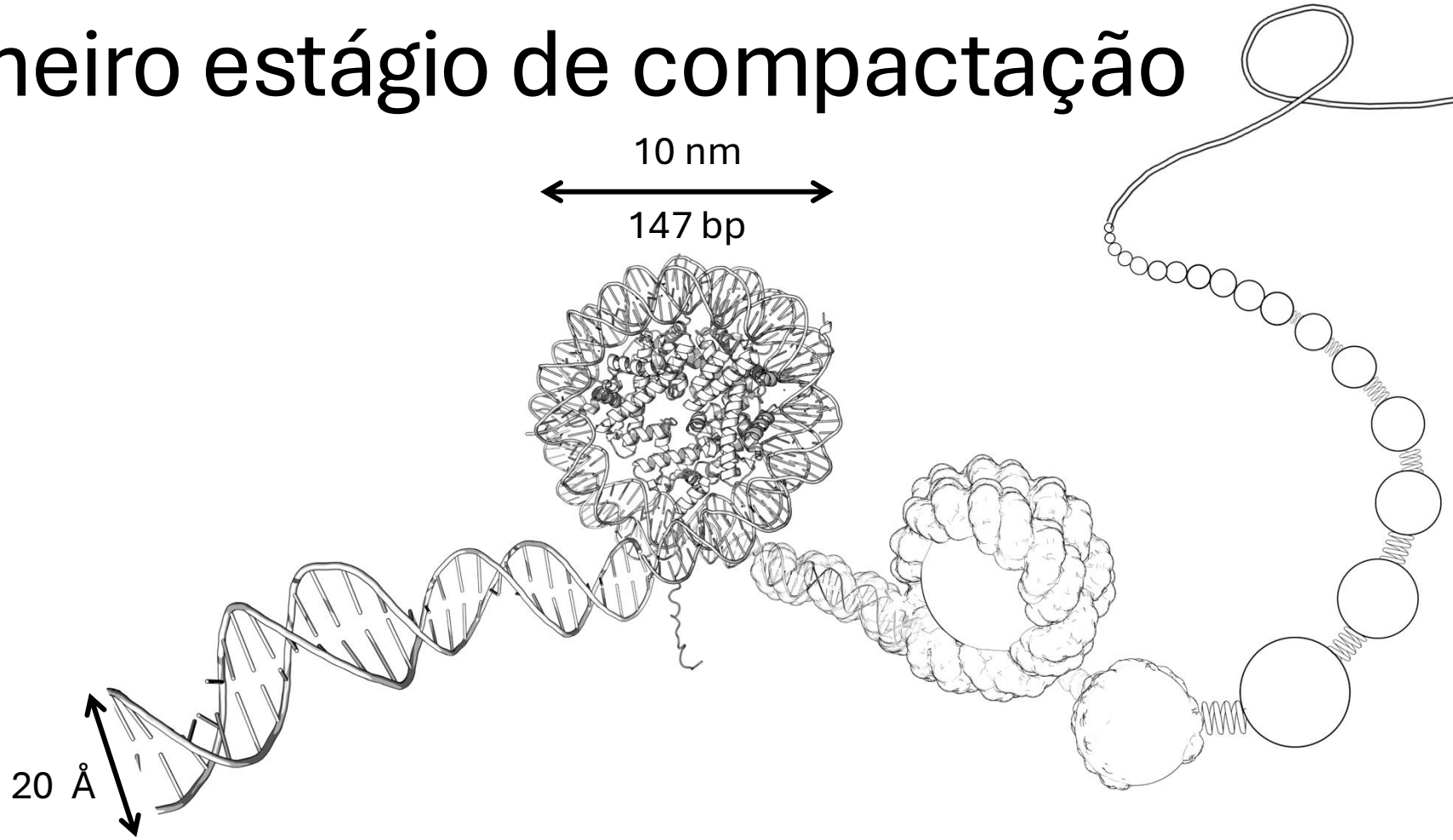


O DNA cobre escalas amplas



Marti-Renom & Mirny , 2011.

Primeiro estágio de compactação



A fibra de cromatina também é um polímero muito muito longo.

Objeto	Espessura	Comprimento
1 Mbp DNA	20 Å	0,34 mm
1 Mbp cromatina	10 nm	50 µm

5 000 vezes mais longo que seu diâmetro!

Local x global chromatin scale

At the local scale, the specific sequence matters.

Local x global chromatin scale

„In a global view (…) we extract simple, universal features which will remain true for a large class of polymer chains.“

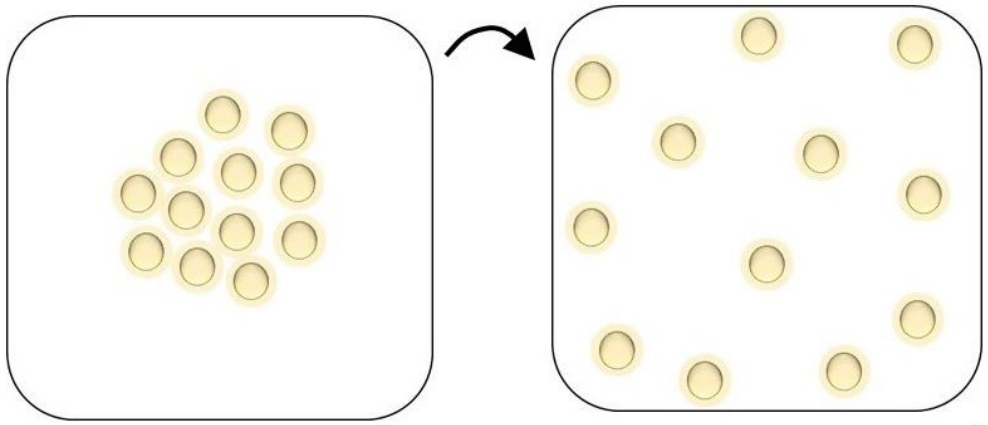
Pierre-Gilles de Gennes,
Physics Nobel Prize 1991

Universal properties of long polymers

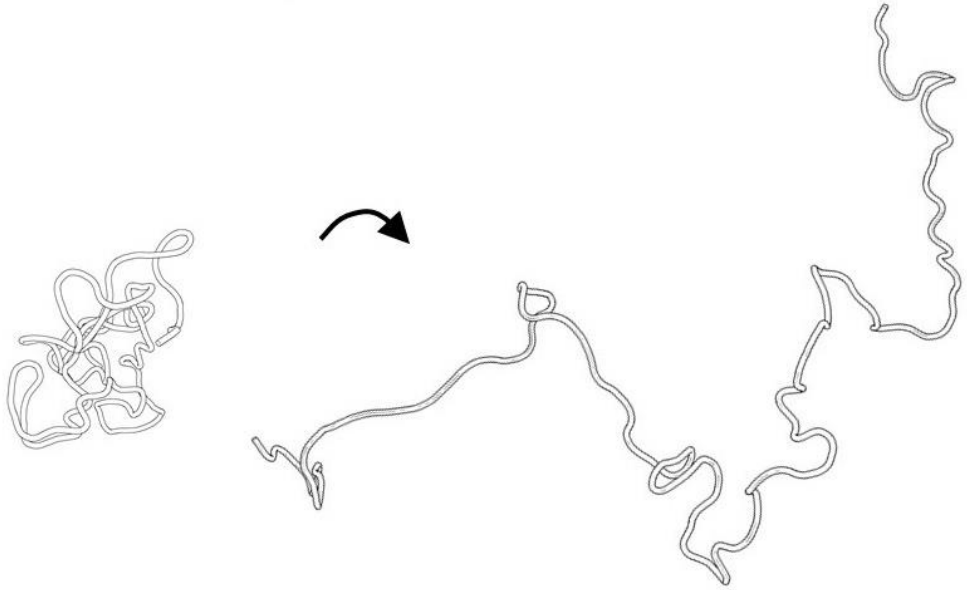
- Subdiffusion
- Depletion attraction
- Lack of reptation

Subdiffusion

Diffusion of particles

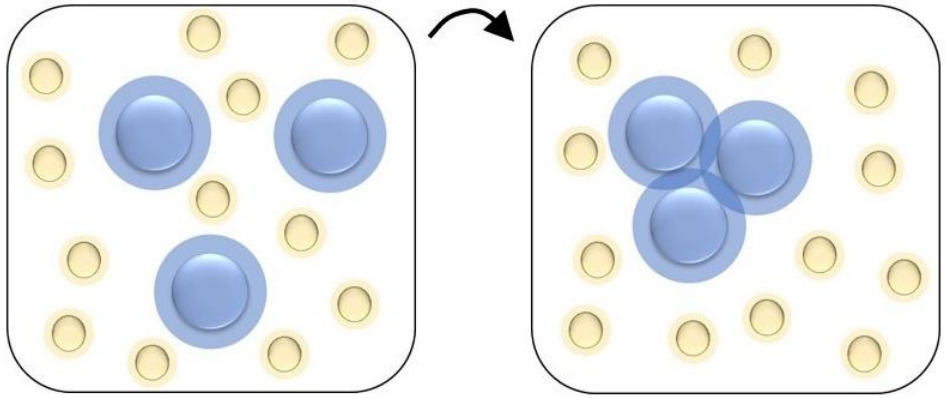


Diffusion of polymers

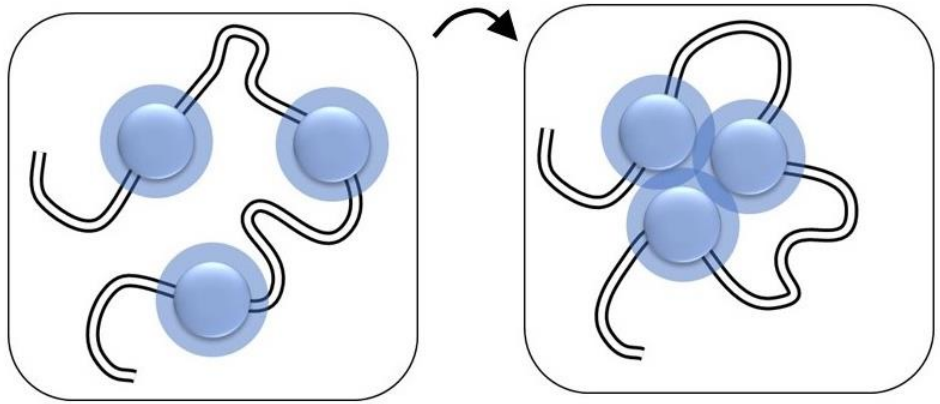


Depletion attraction

Excluded volume between particles



Excluded volumes with a polymer



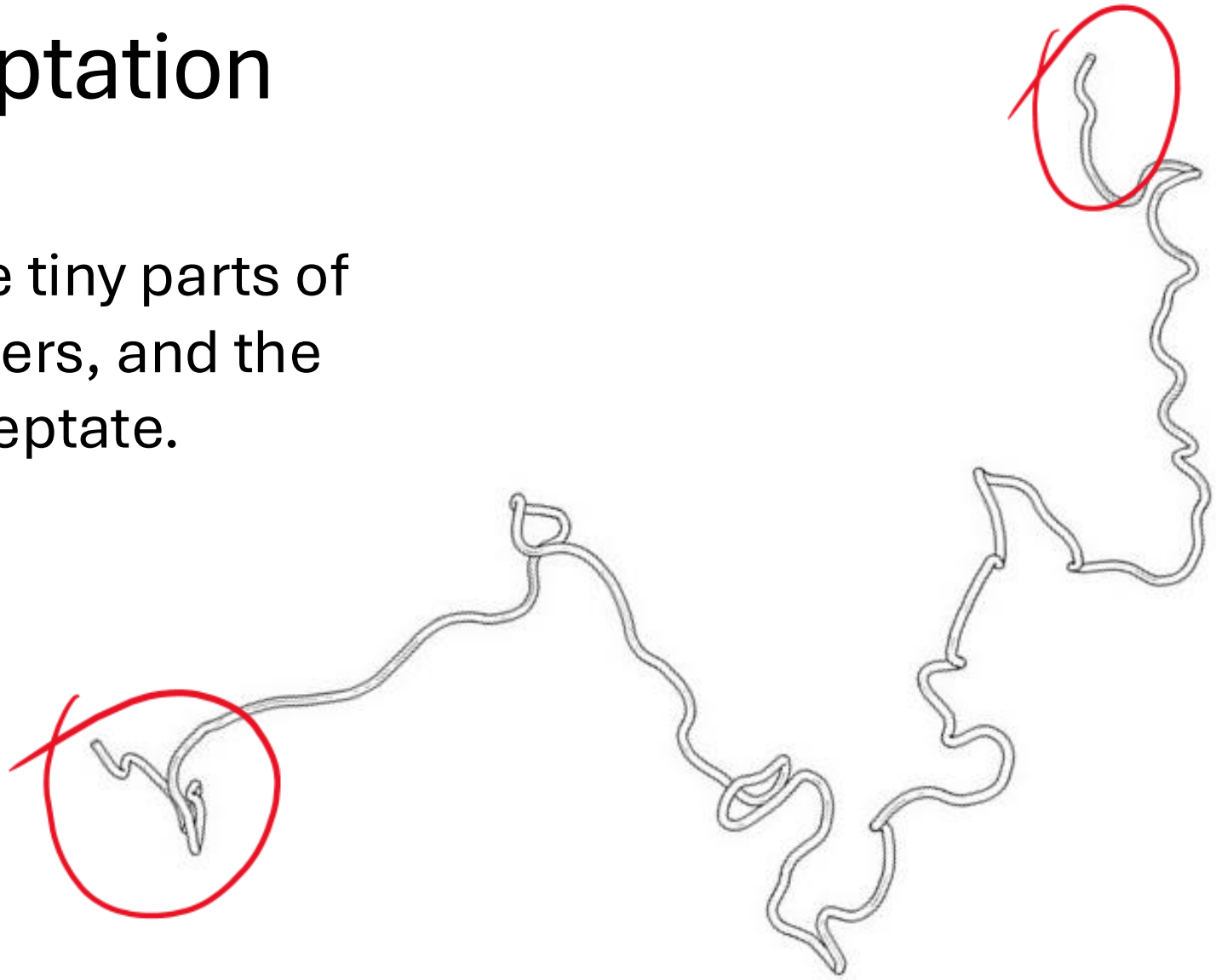
Reptation





Lack of reptation

The tips are tiny parts of long polymers, and the rest can’t reptate.

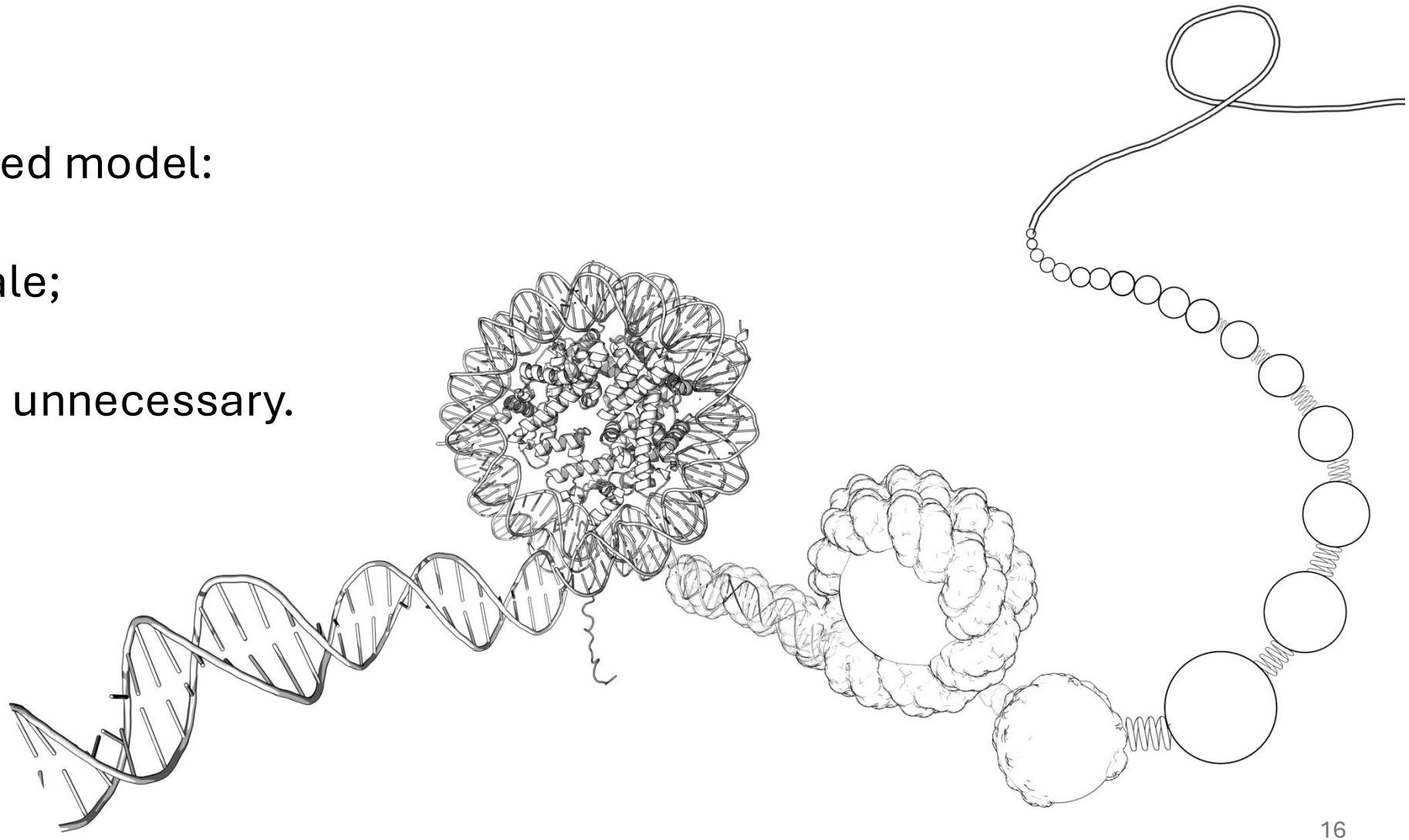


How to consider these effects in computer models?

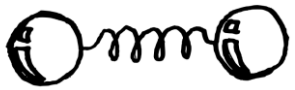
A chromatin model

Coarse-grained model:

1. Global scale;
2. Low cost;
3. Avoids the unnecessary.

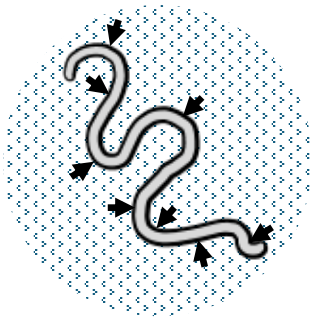


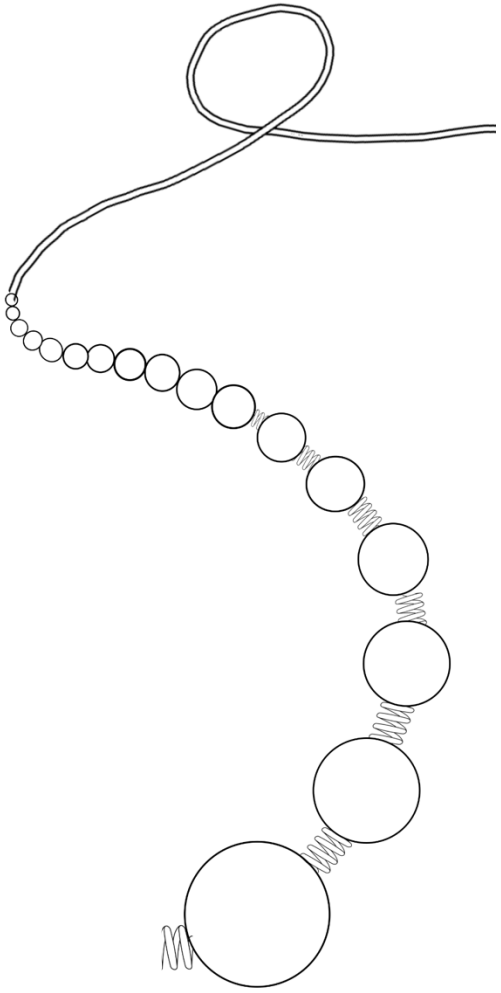
Physical forces between the monomers

1. 

2. 

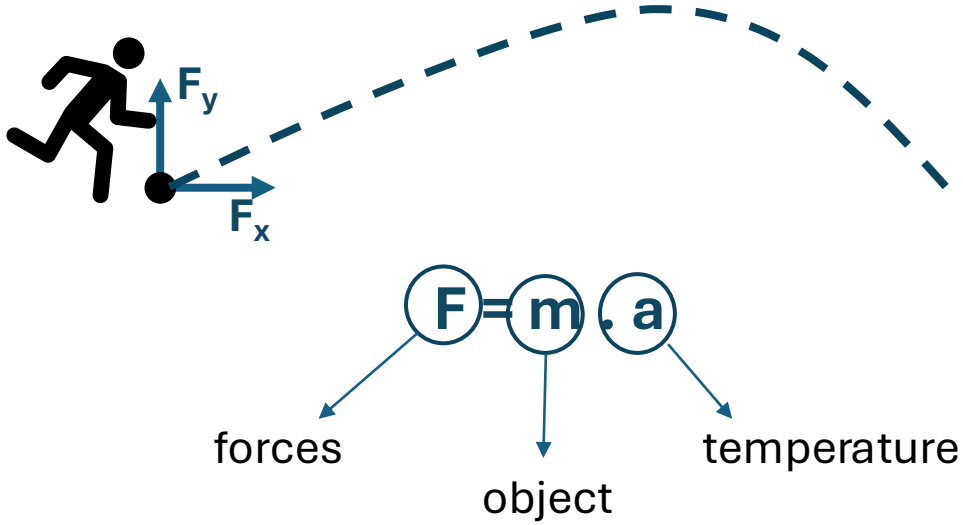
3. 

4. 

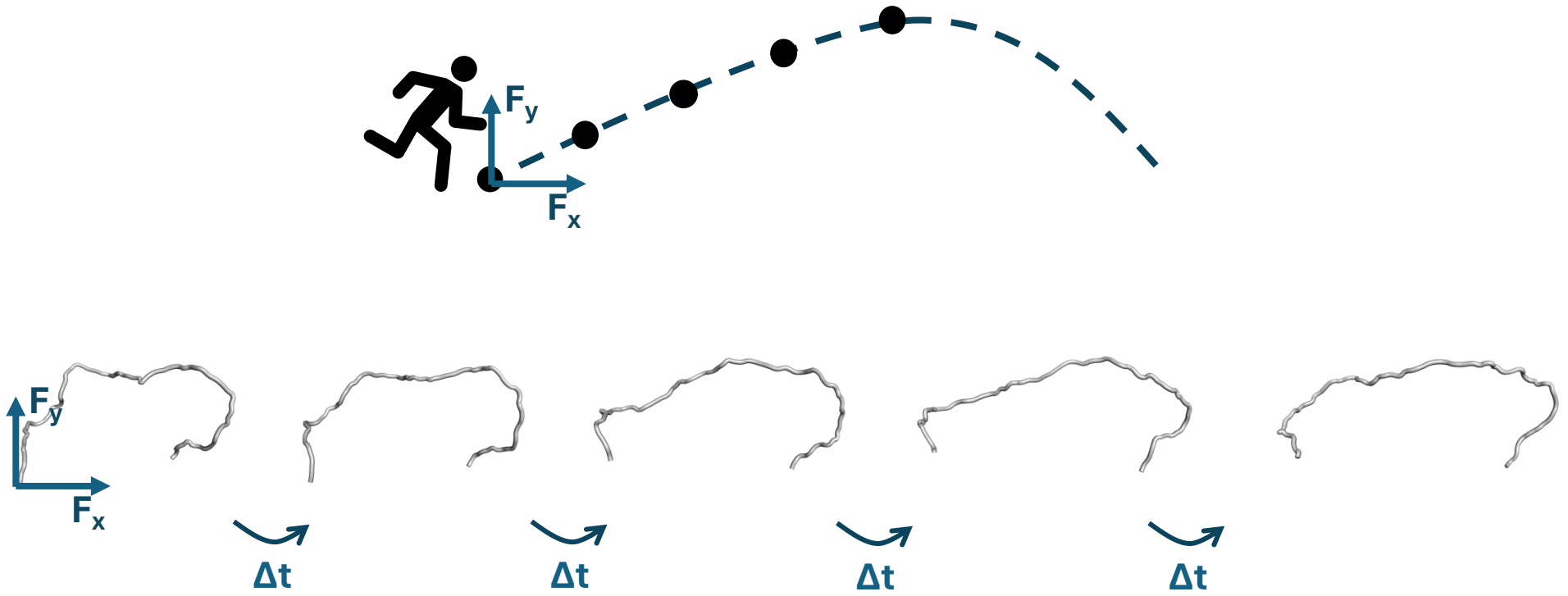


https://youtu.be/rwMXRvGDIyA

… and a physics approach

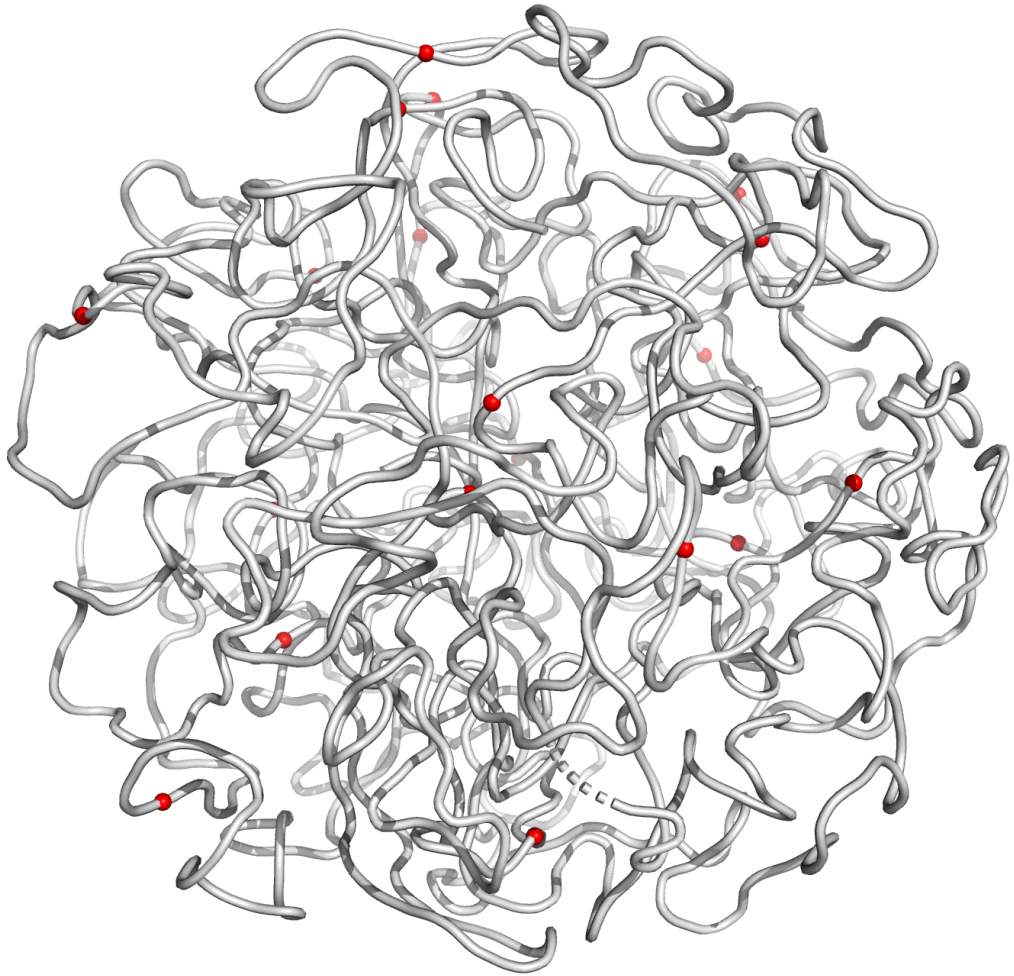


The heating

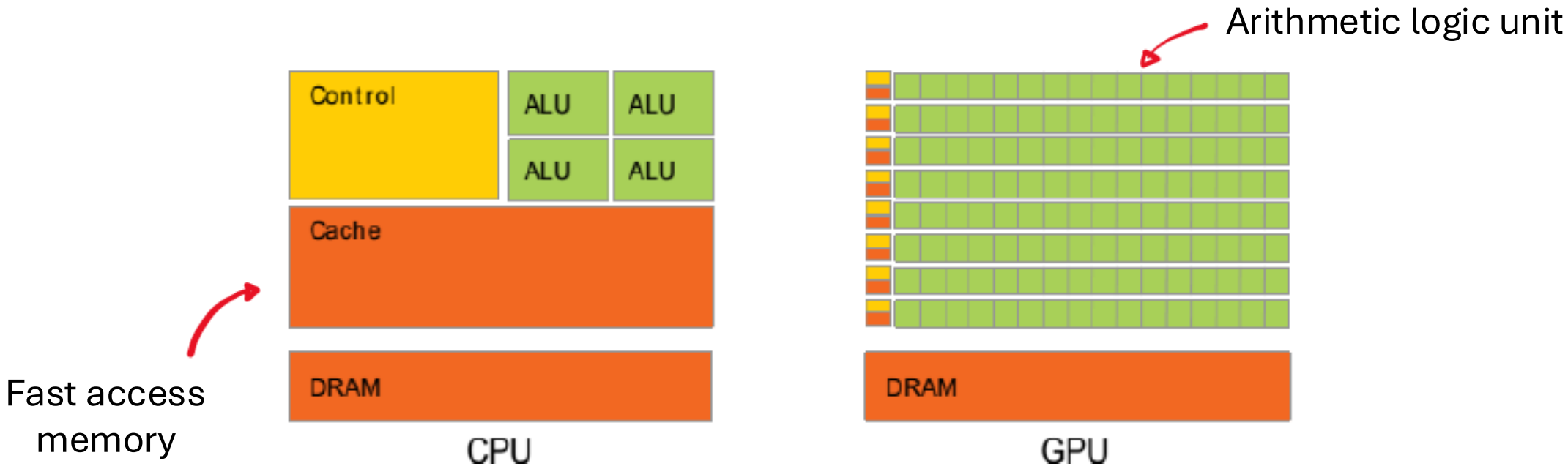


Finally, a trajectory.
A polymer that moves like chromatin!

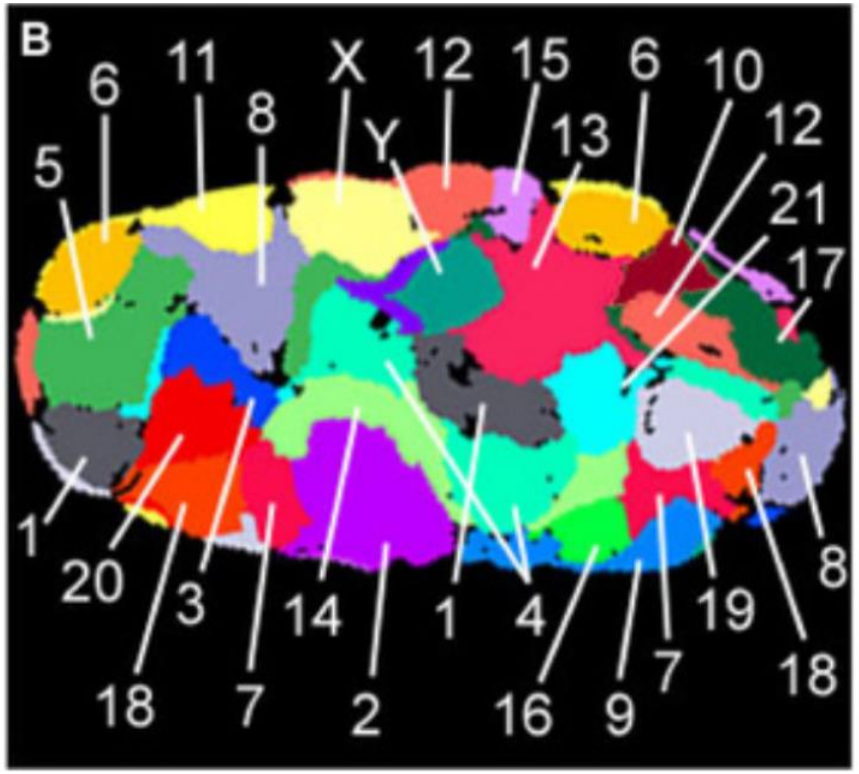
A fiber that moves like chromatin



High performance computing
Graphics processing unit (GPU)

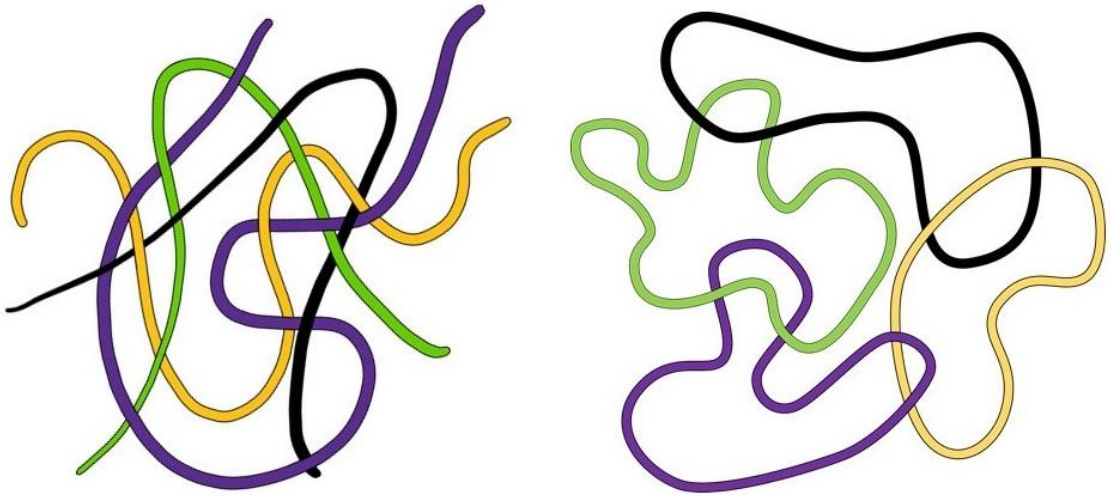


Simulation of chromosome territories



Human fibroblast

Bolzer et al., 2005

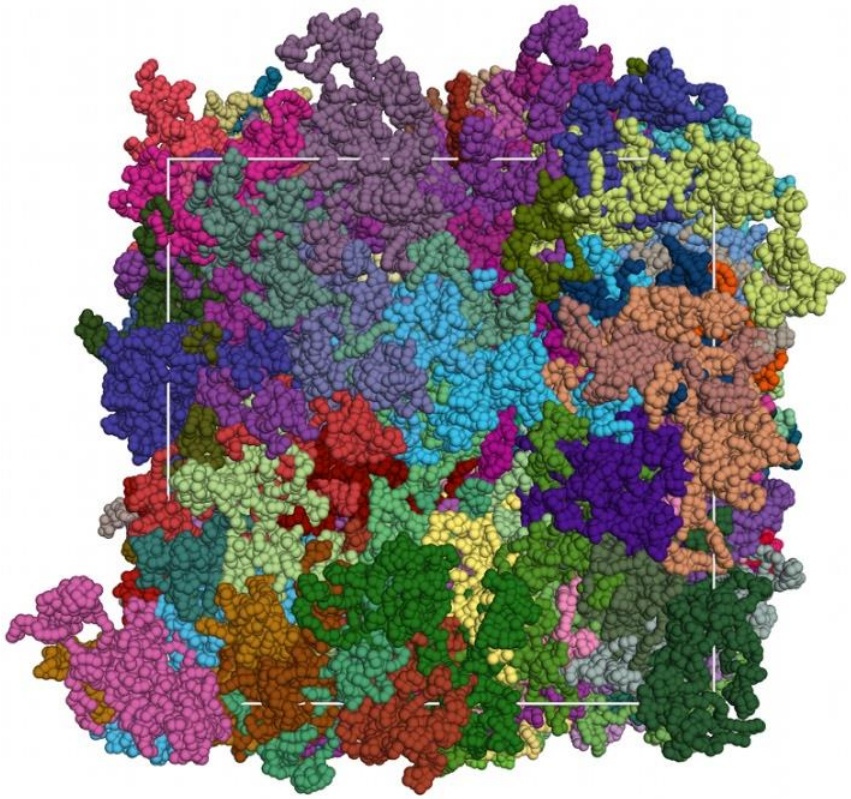




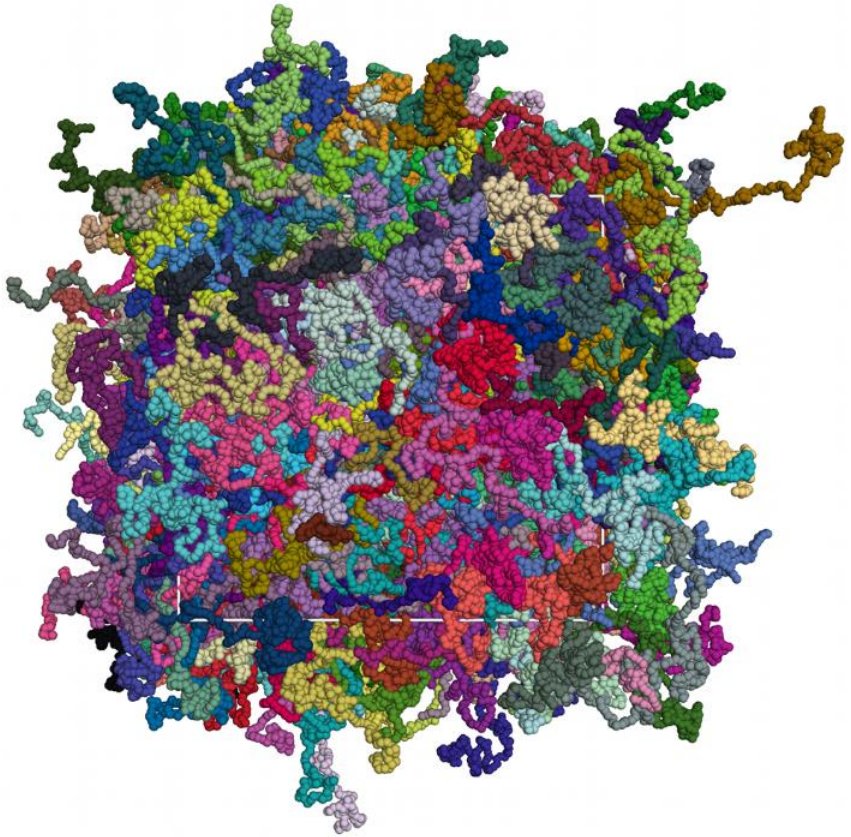
Halverson et al., 2014

https://www.youtube.com/watch?v=B_t8tzrtyYI

Simulation of chromosome territories



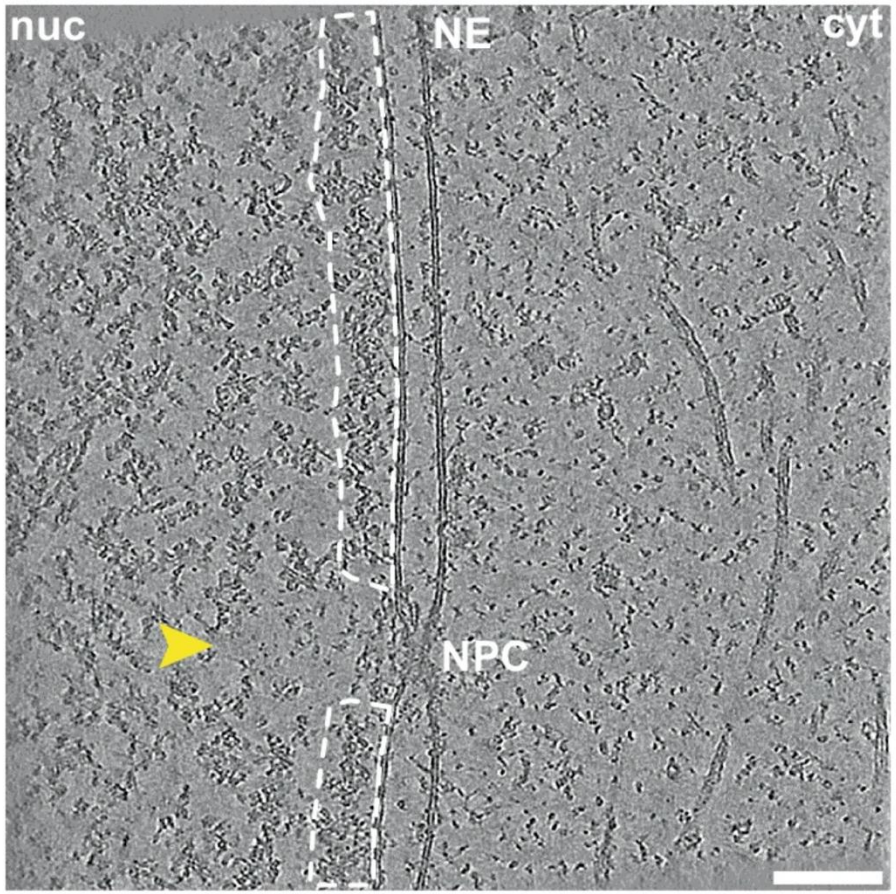
Polymer rings



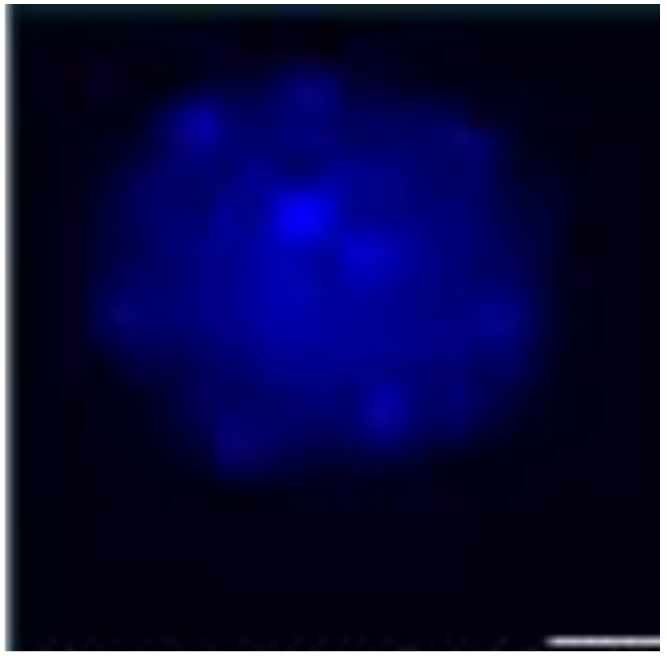
Linear polymers

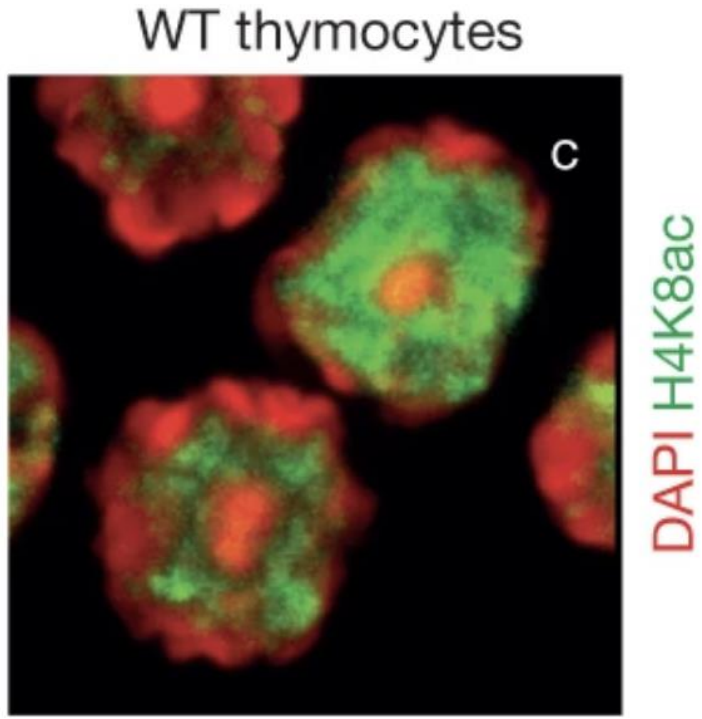
Halverson et al., 2014

Heterocromatina e eucromatina



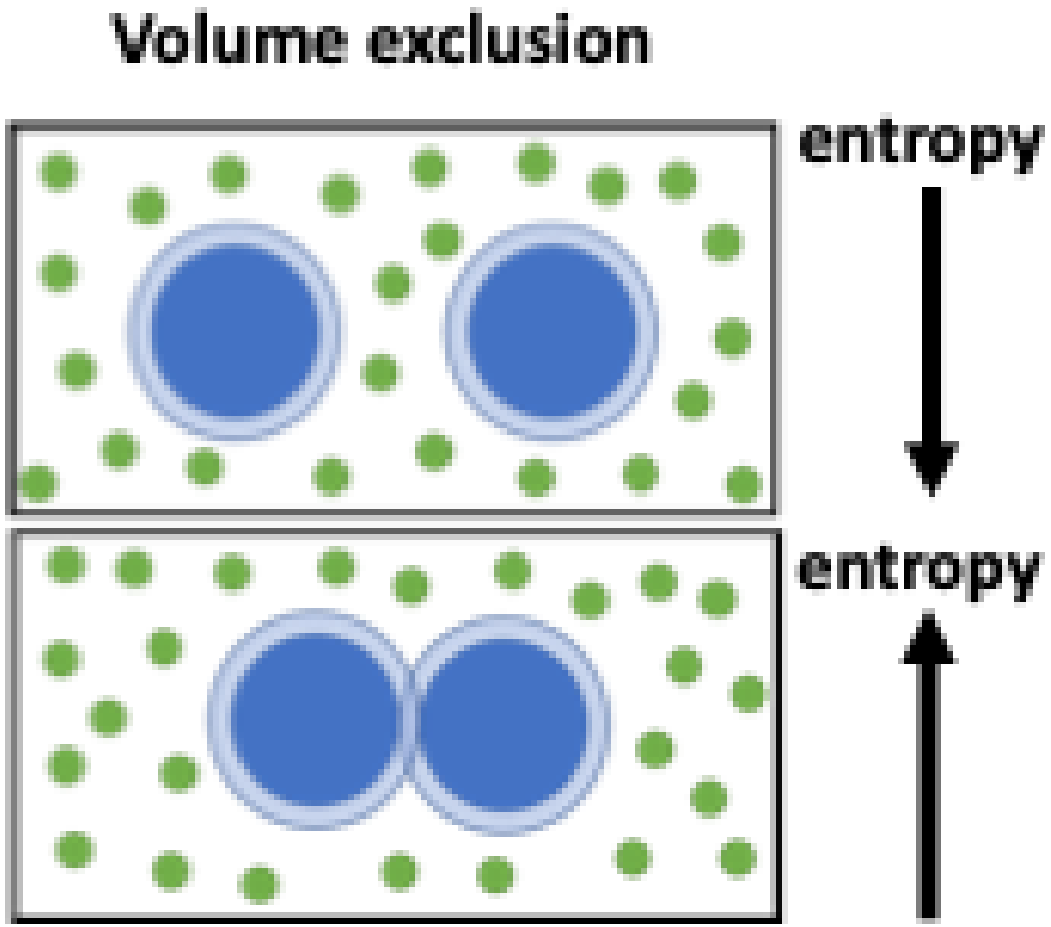
Kreysing et al., 2026

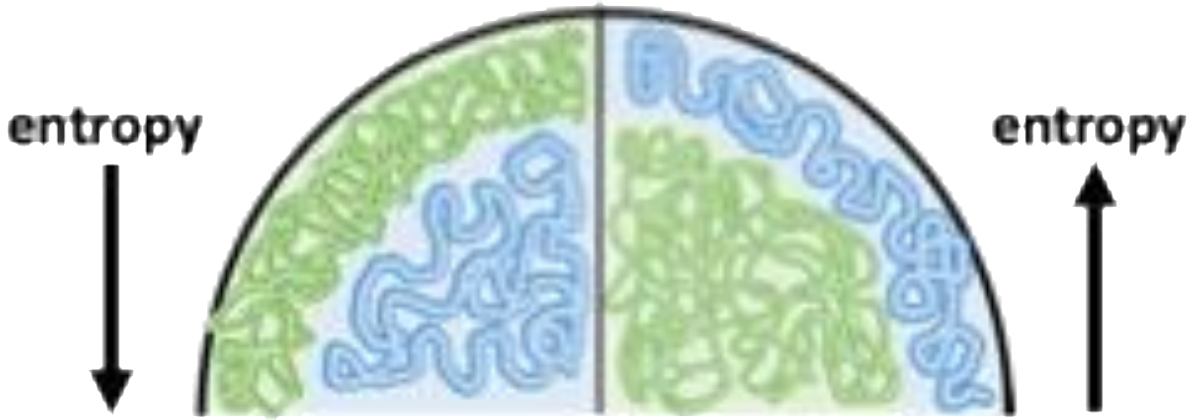




Falk et al., 2019

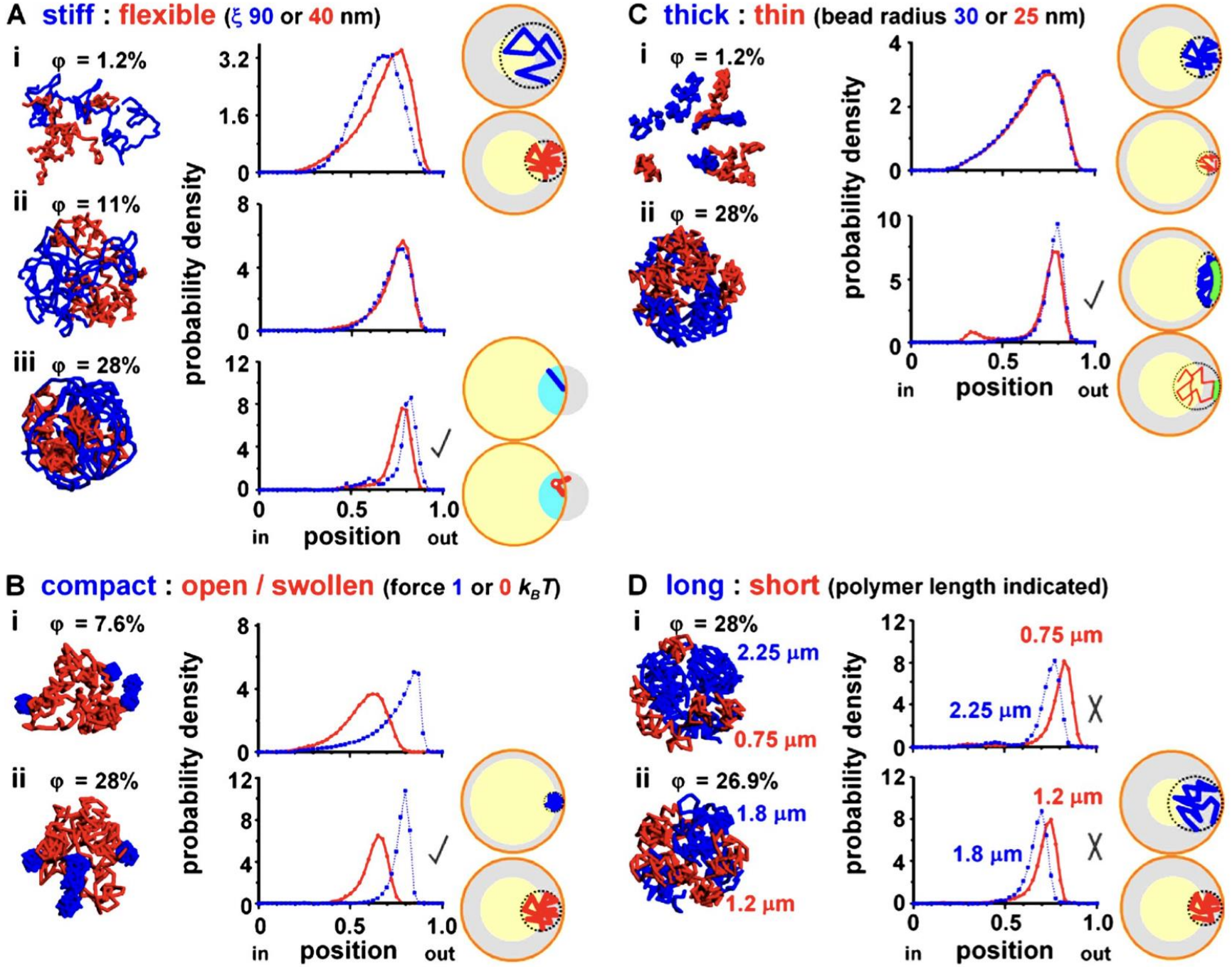
Separação de fase



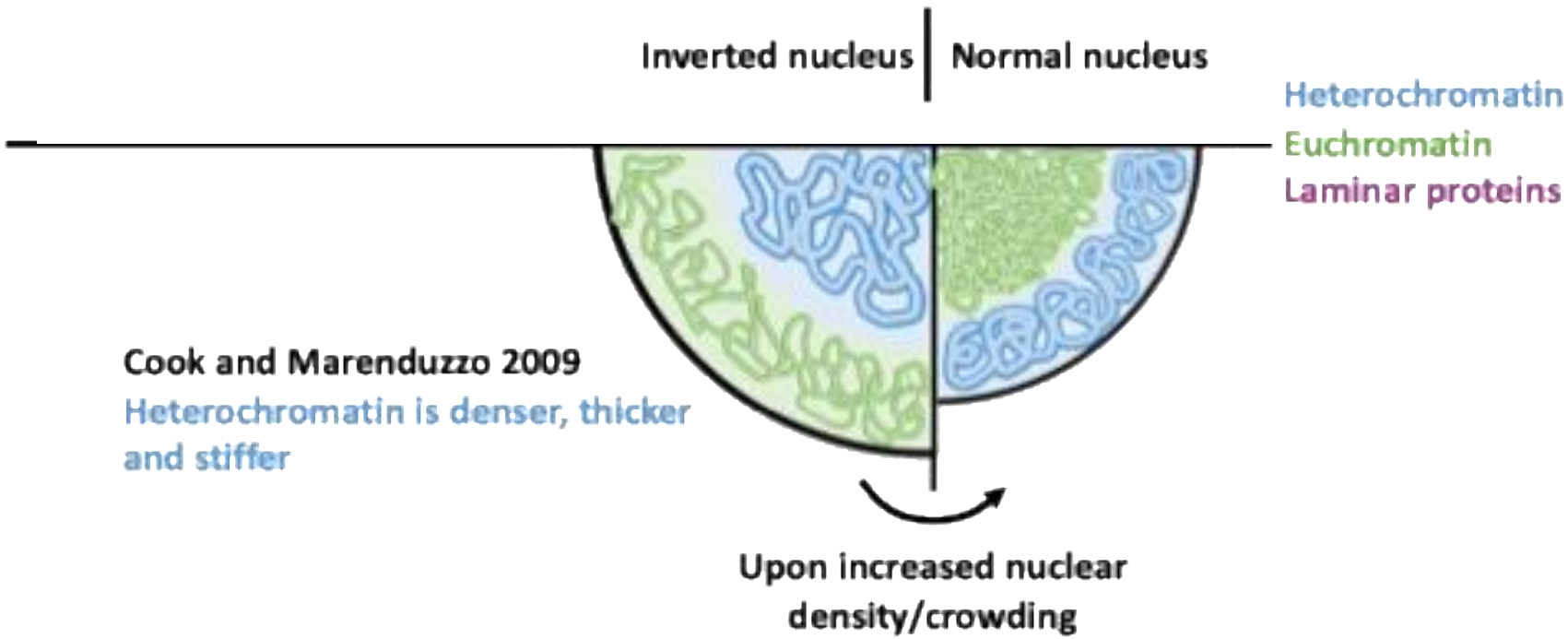


Camara and Mascher , 2023

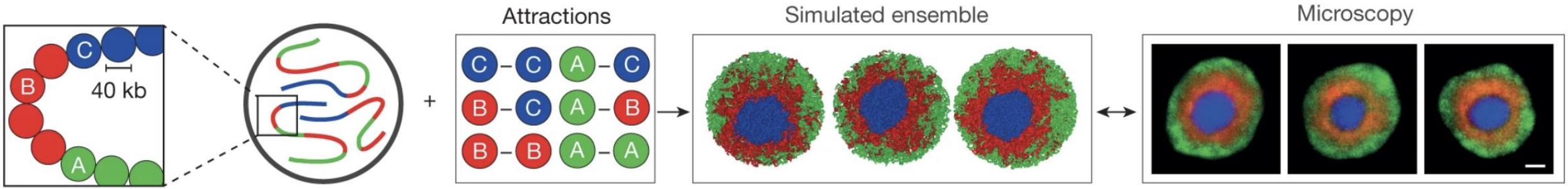
Mudando características dos nucleossomos

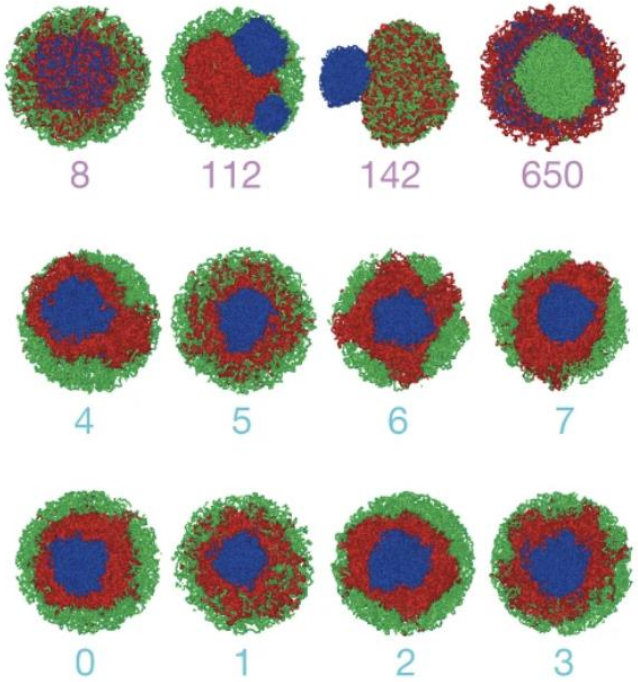


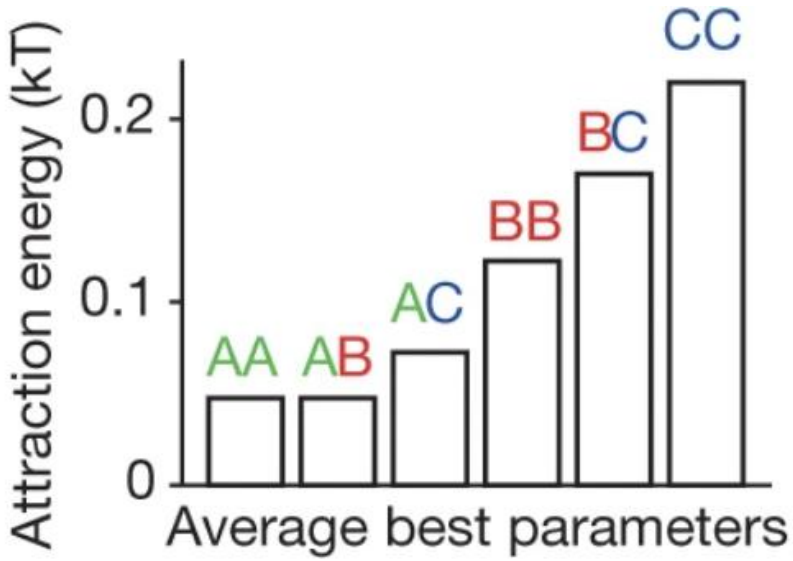
Cook and Marenduzzo, 2009

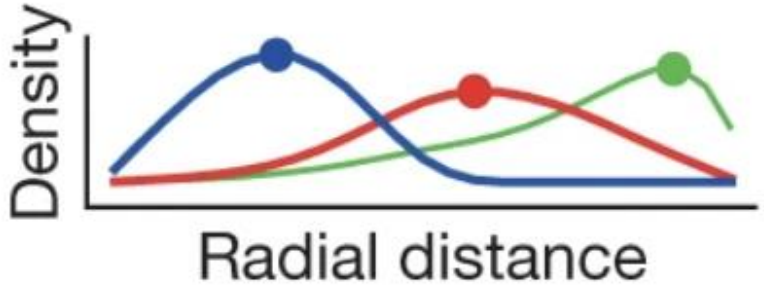

Inverted nucleus
Normal nucleus
Heterochromatin
Euchromatin
Laminar proteins
Cook and Marenduzzo 2009
Heterochromatin is denser, thicker and stiffer
Upon increased nuclear density/crowding

A – eucromatina
B – heterocromatina
C – repeats, regiões pericentromérica



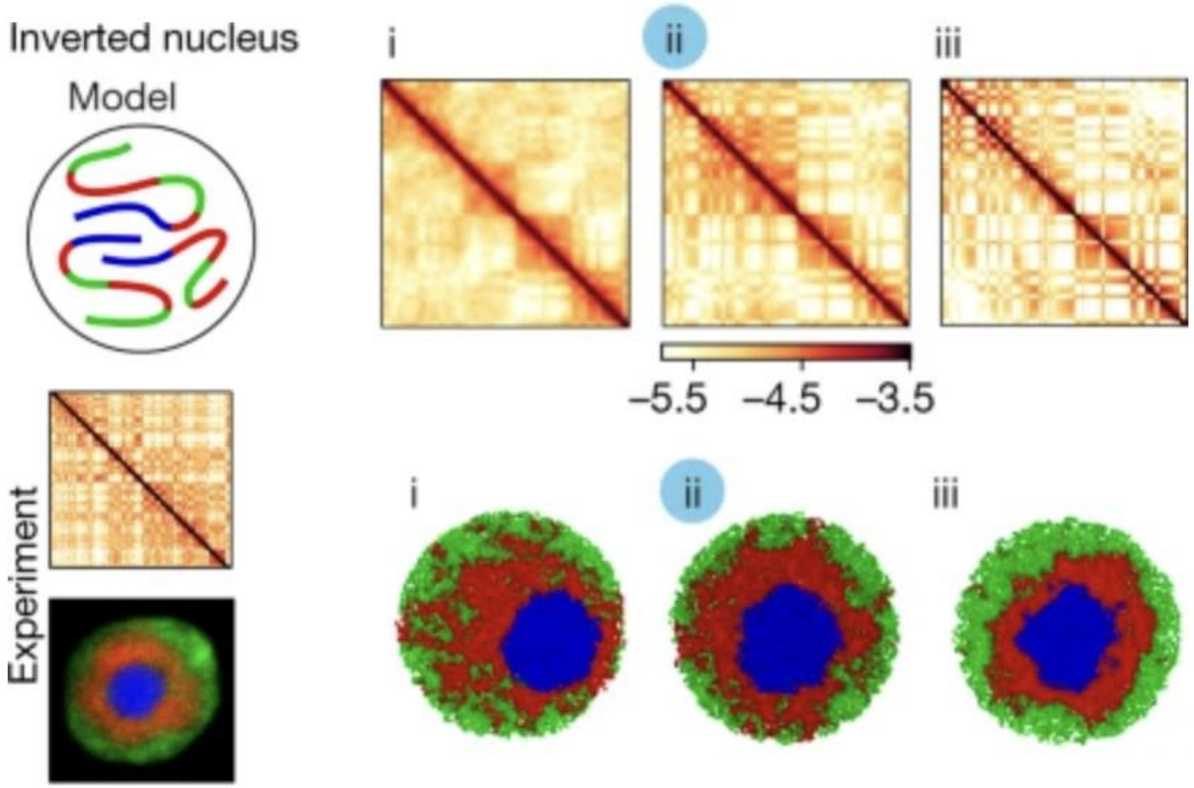




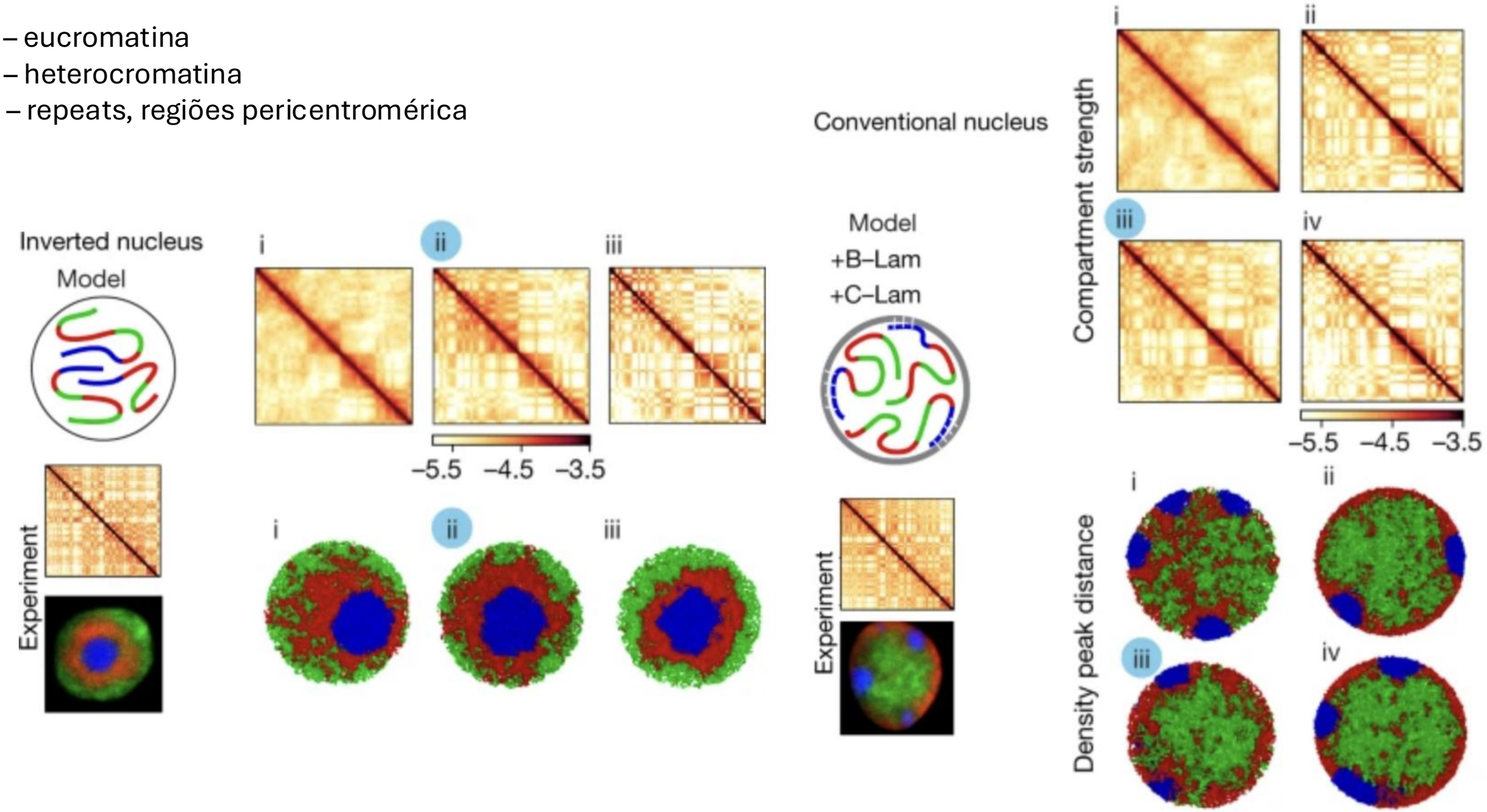


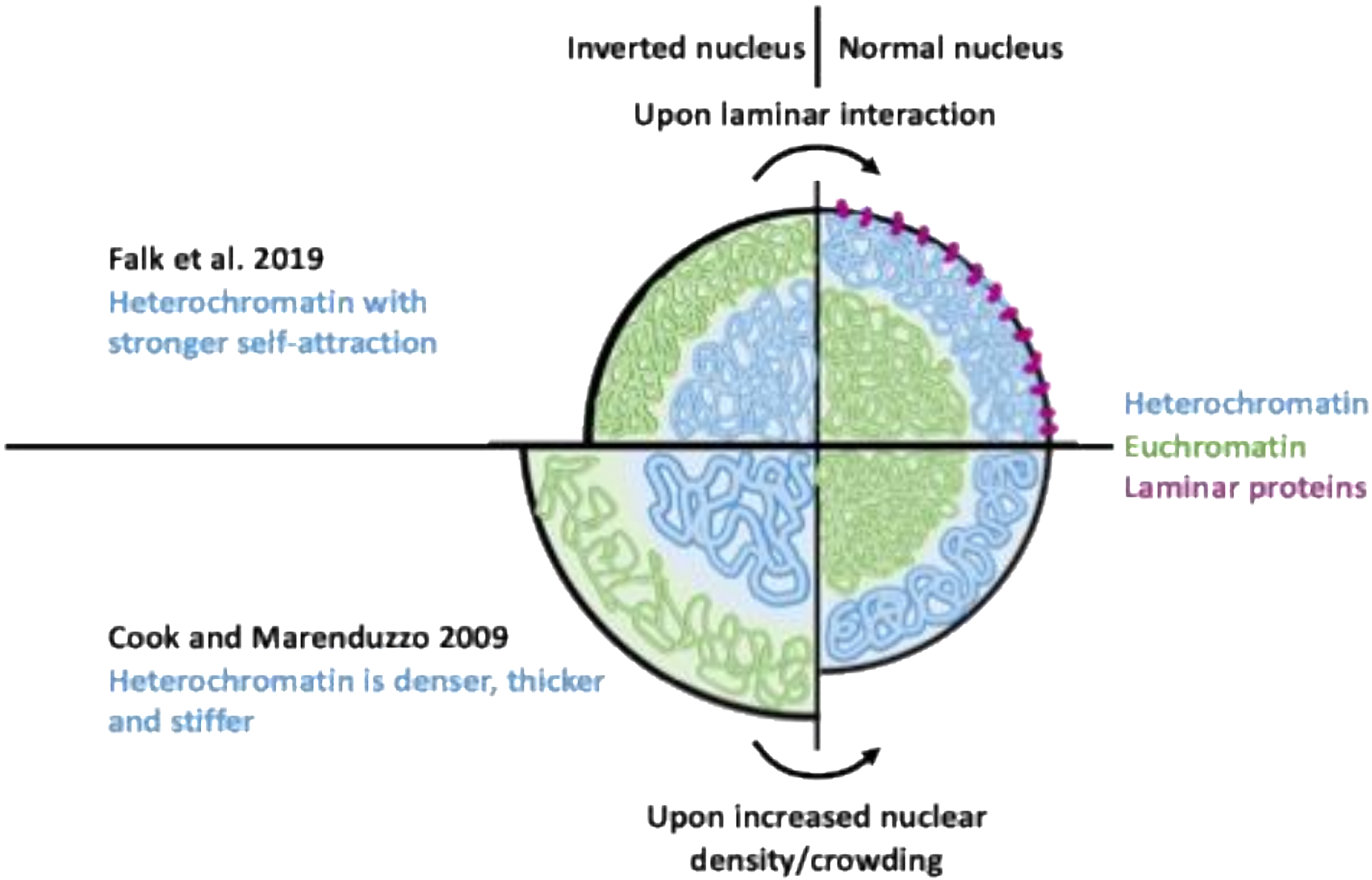
Falk, 2019

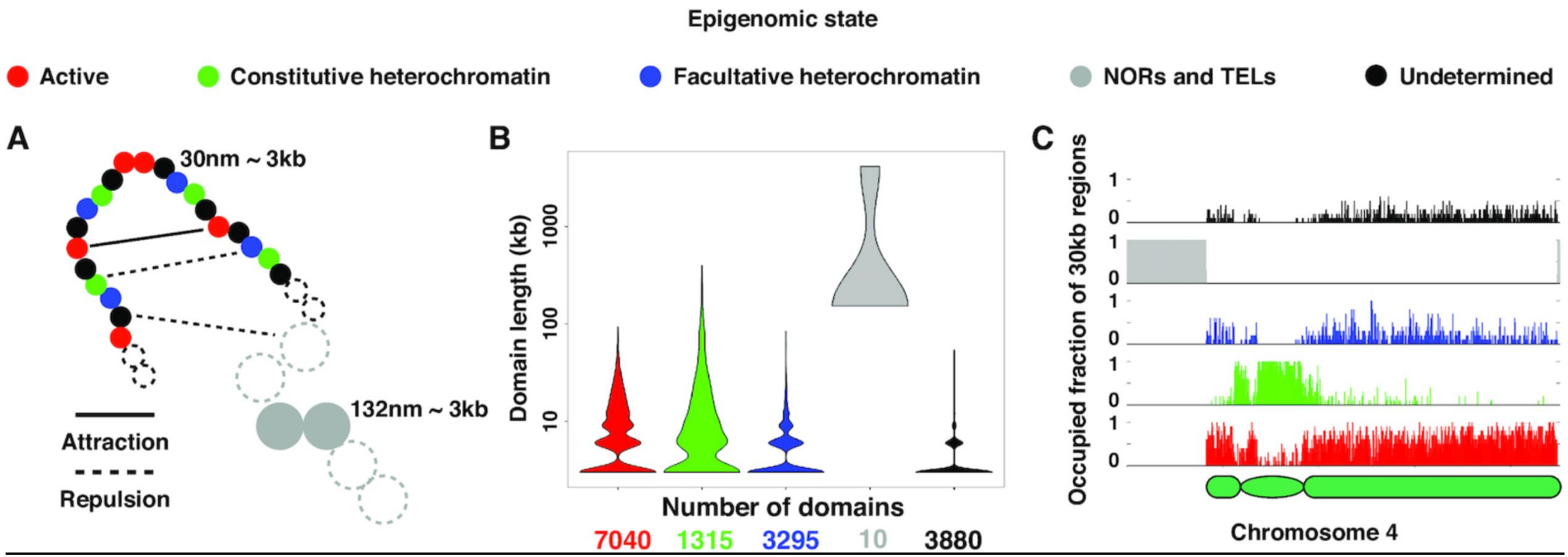
A – eucromatina
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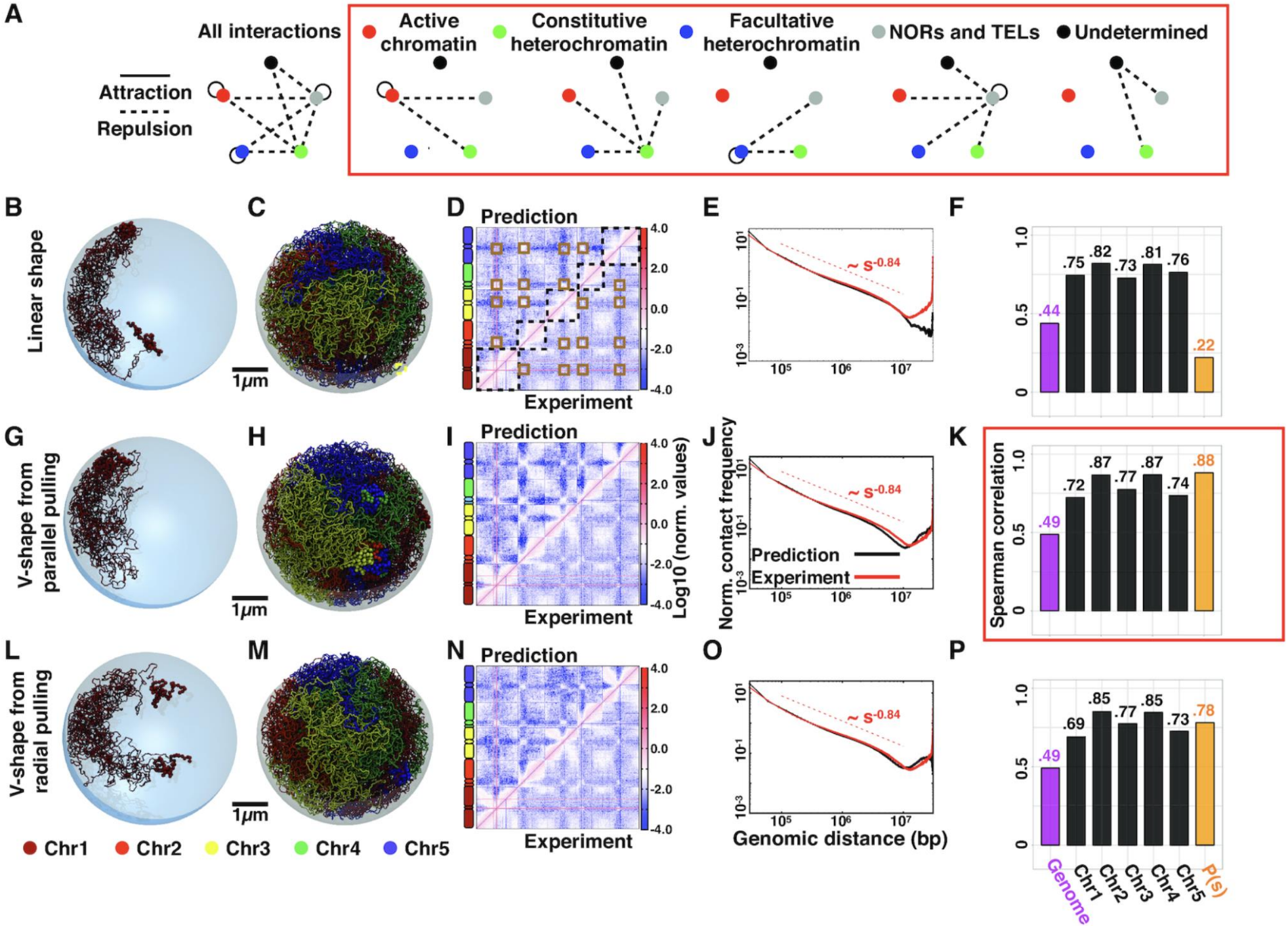


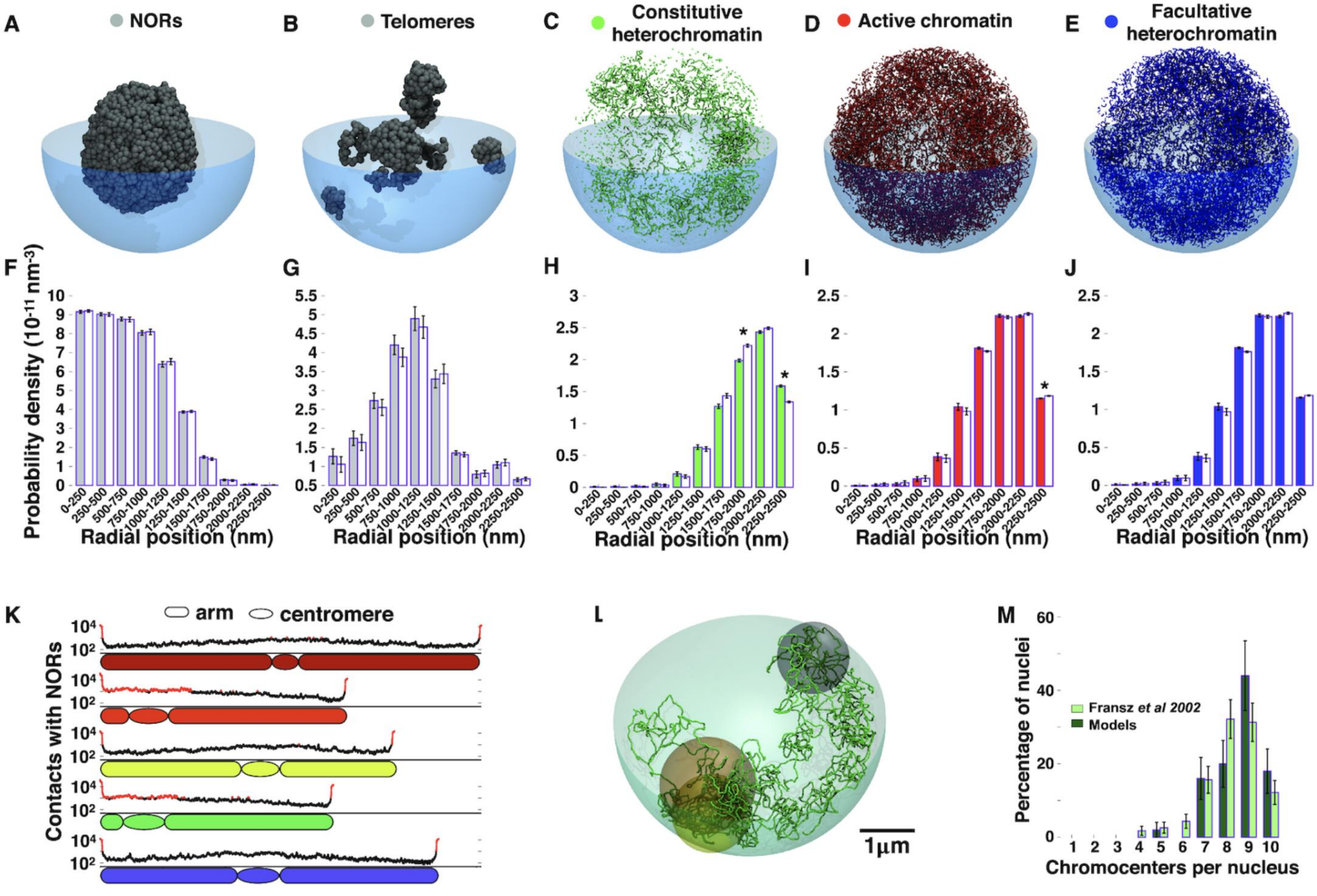
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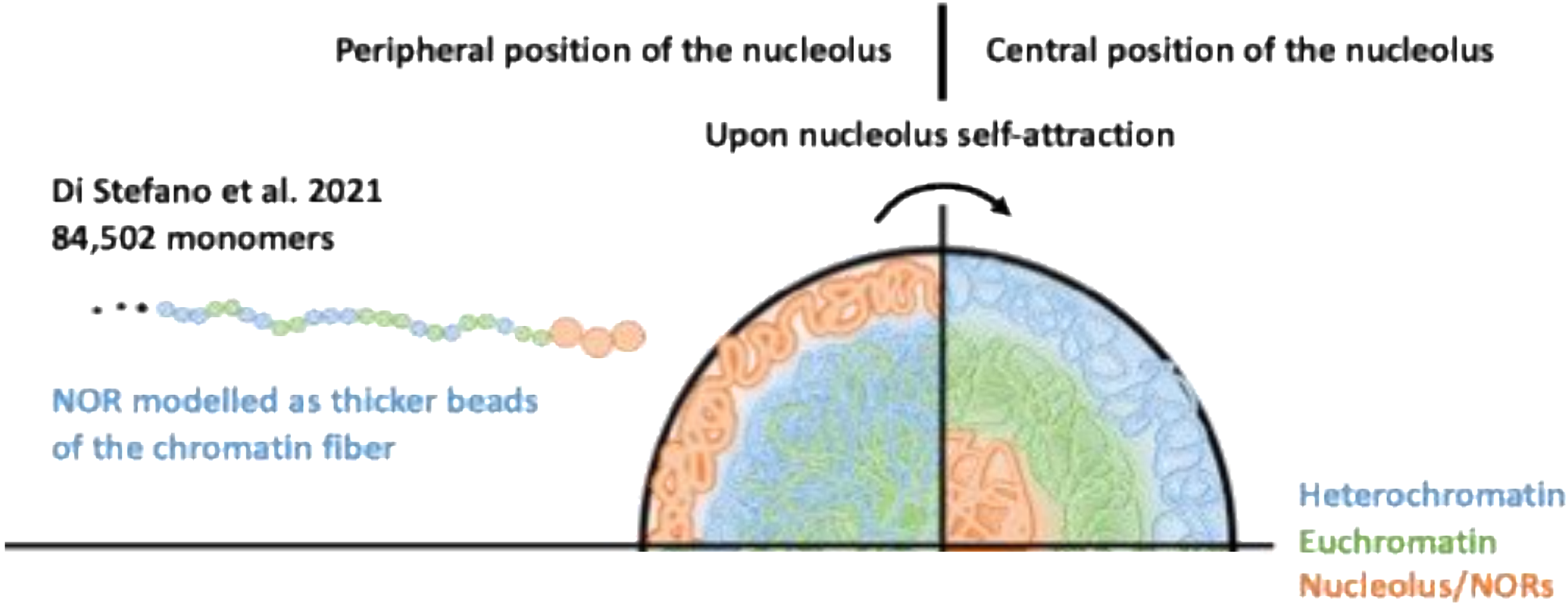


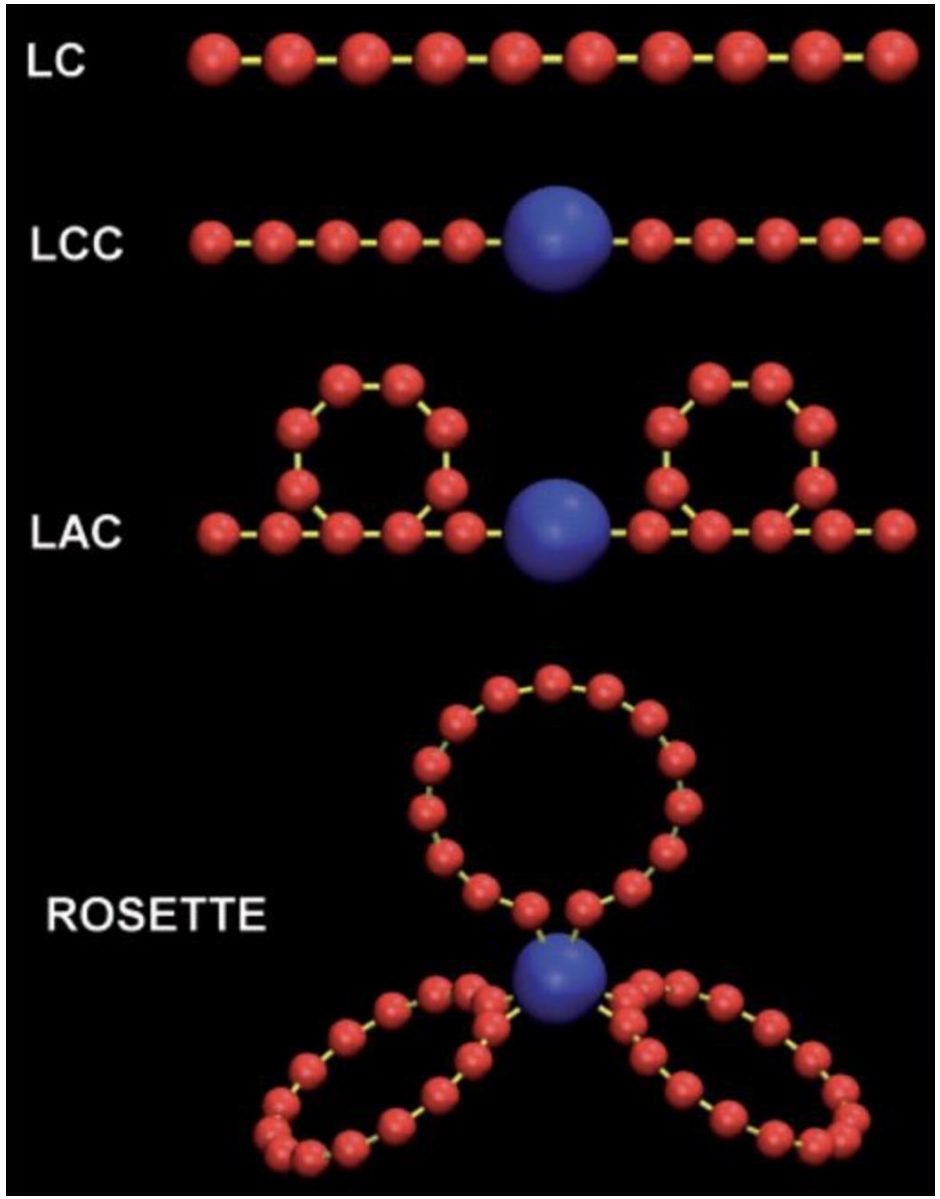

Inverted nucleus
Normal nucleus
Upon laminar interaction
Falk et al. 2019
Heterochromatin with stronger self-attraction
Heterochromatin
Euchromatin
Laminar proteins
Cook and Marenduzzo 2009
Heterochromatin is denser, thicker and stiffer
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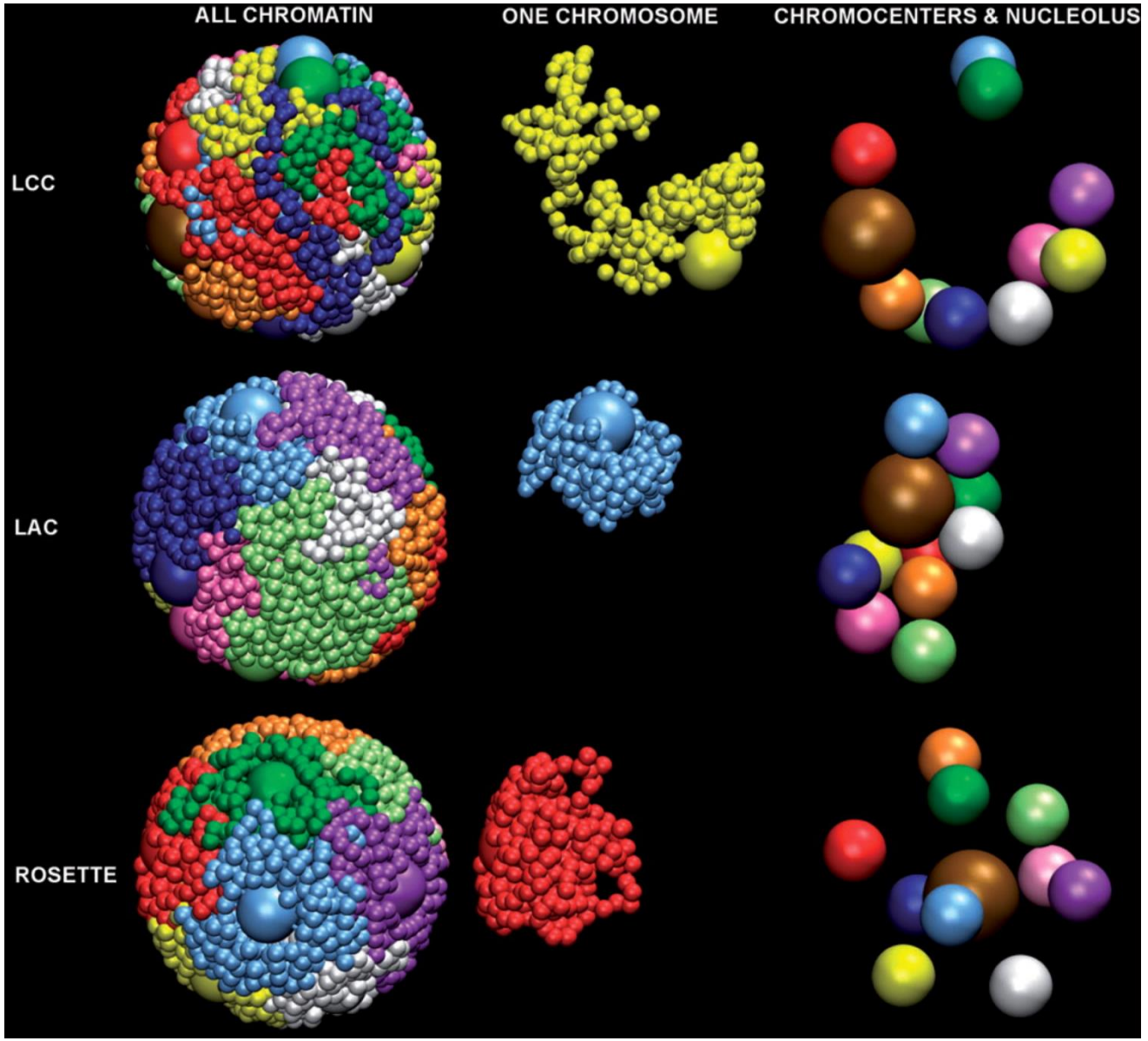

Epigenomic state
Active
Constitutive heterochromatin
Facultative heterochromatin
NORs and TELs
Undetermined
A
30nm ~ 3kb
132nm ~ 3kb
Attraction
Repulsion
B
Domain length (kb)
1000
100
10
Number of domains
7040
1315
3295
10
3880
C
Occupied fraction of 30kb regions
1
0
Chromosome 4


A
All interactions
Attraction
Repulsion
Active chromatin
Constitutive heterochromatin
Facultative heterochromatin
NORs and TELs
Undetermined
B
Linear shape
C
1μm
D
Prediction
Experiment
E
~ s-0.84
F
.44
.75
.82
.73
.81
.76
.22
G
V-shape from parallel pulling
H
1μm
I
Prediction
Experiment
Log10 (norm. values)
J
Norm. contact frequency
~ s-0.84
Prediction
Experiment
K
Spearman correlation
.49
.72
.87
.77
.87
.74
.88
L
V-shape from radial pulling
M
1μm
N
Prediction
Experiment
O
~ s-0.84
Genomic distance (bp)
P
.49
.69
.85
.77
.85
.73
.78
Genome
Chr1
Chr2
Chr3
Chr4
Chr5
P(s)
Chr1
Chr2
Chr3
Chr4
Chr5

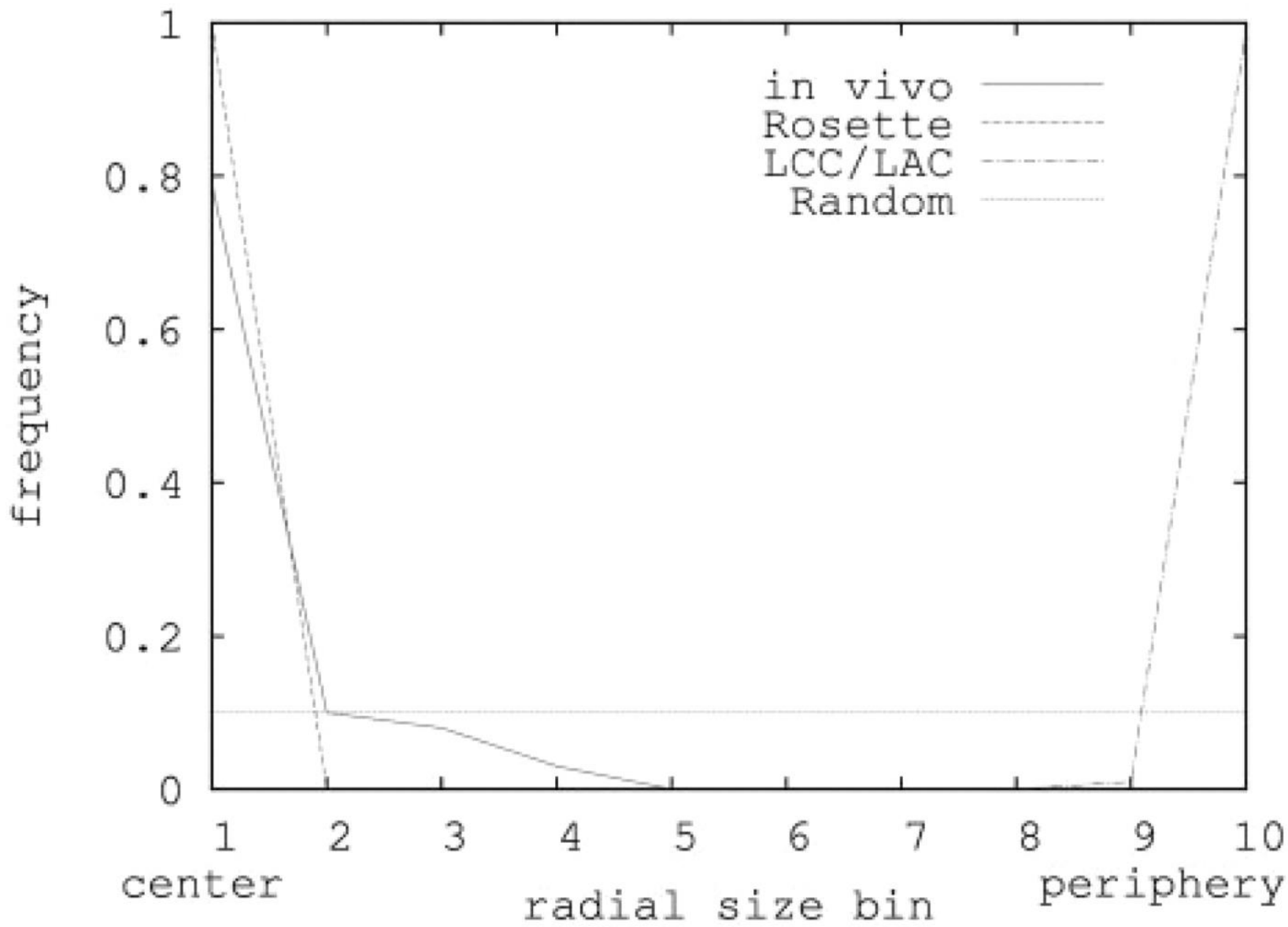

A
NORs
B
Telomeres
C
Constitutive heterochromatin
D
Active chromatin
E
Facultative heterochromatin
F
G
H
I
J
Probability density (10-11 nm-3)
Radial position (nm)
0-250
250-500
500-750
750-1000
1000-1250
1250-1500
1500-1750
1750-2000
2000-2250
2250-2500
K
arm
centromere
Contacts with NORs
L
1μm
M
Percentage of nuclei
Fransz et al 2002
Models
Chromocenters per nucleus

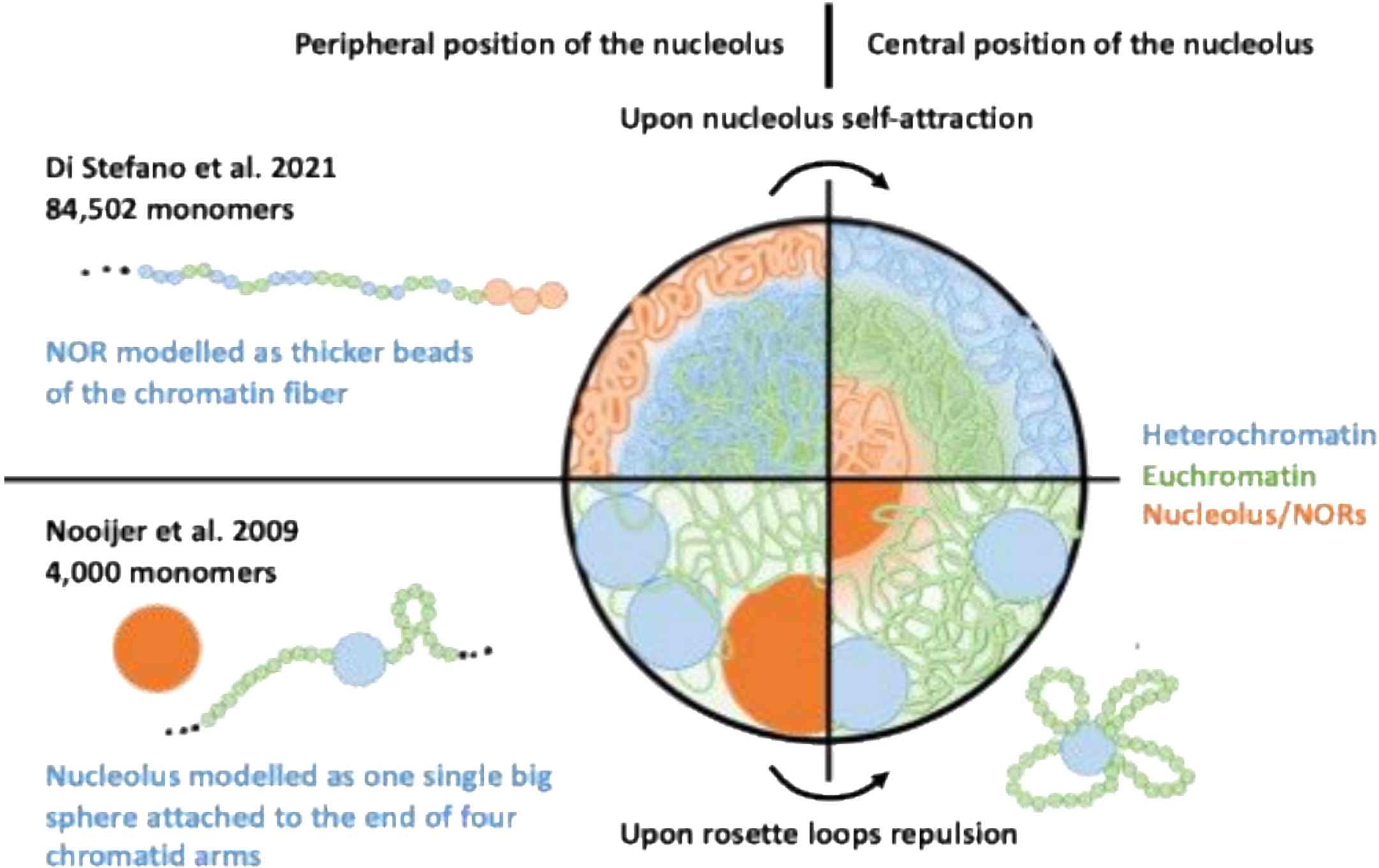

Peripheral position of the nucleolus
Central position of the nucleolus
Upon nucleolus self-attraction
Di Stefano et al. 2021
84,502 monomers
NOR modelled as thicker beads
of the chromatin fiber
Heterochromatin
Euchromatin
Nucleolus/NORs


LC
LCC
LAC
ROSETTE


ALL CHROMATIN
ONE CHROMOSOME
CHROMOCENTERS & NUCLEOLUS
LCC
LAC
ROSETTE

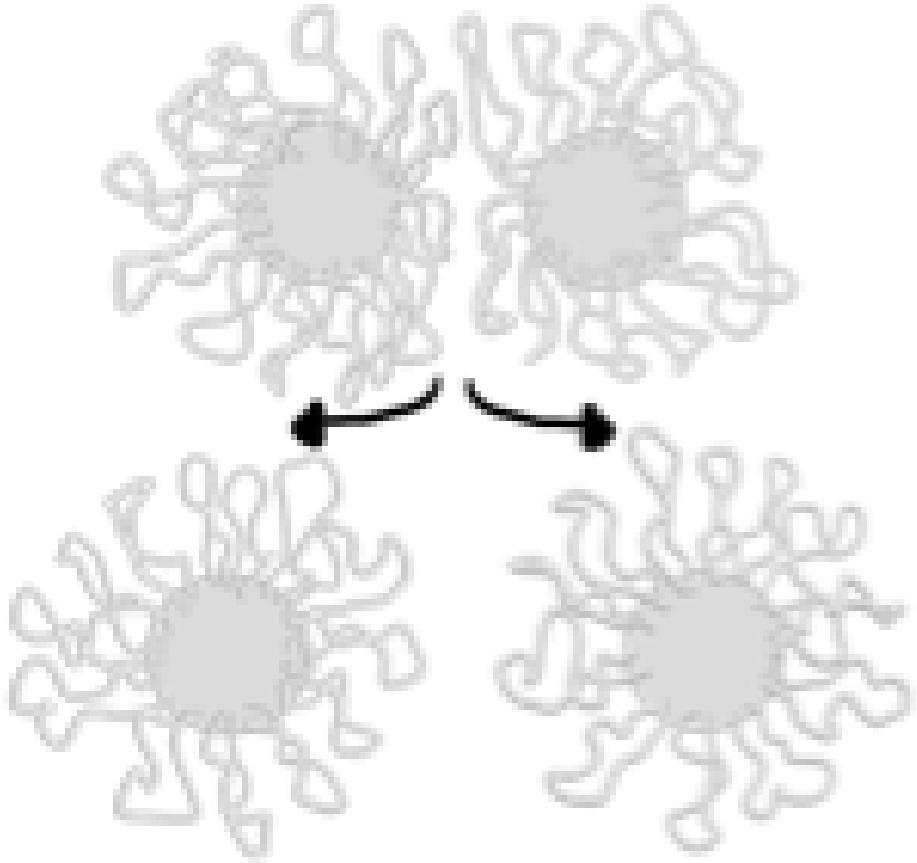
Posição do nucleolus

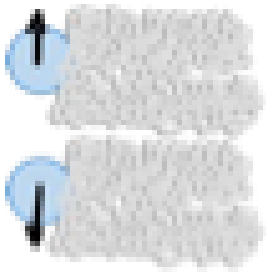



Peripheral position of the nucleolus
Central position of the nucleolus
Upon nucleolus self-attraction
Di Stefano et al. 2021
84,502 monomers
NOR modelled as thicker beads of the chromatin fiber
Heterochromatin
Euchromatin
Nucleolus/NORs
Nooijer et al. 2009
4,000 monomers
Nucleolus modelled as one single big sphere attached to the end of four chromatid arms
Upon rosette loops repulsion

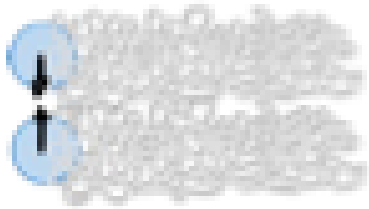
Repulsão de loops

Steric hindrance/osmotic repulsion





Presence of condensin II
High compaction



Depletion of condensin II
Low compaction



Mushroom



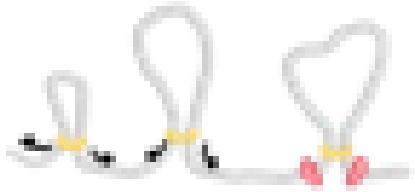
Brush

Como são feitos os loops

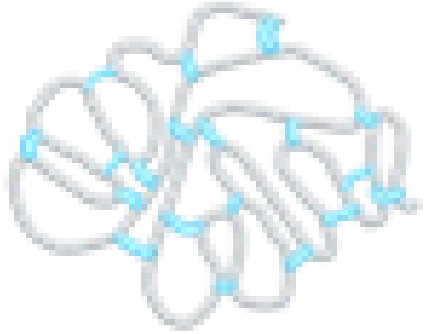


1) Specific loops

2) General self-attraction

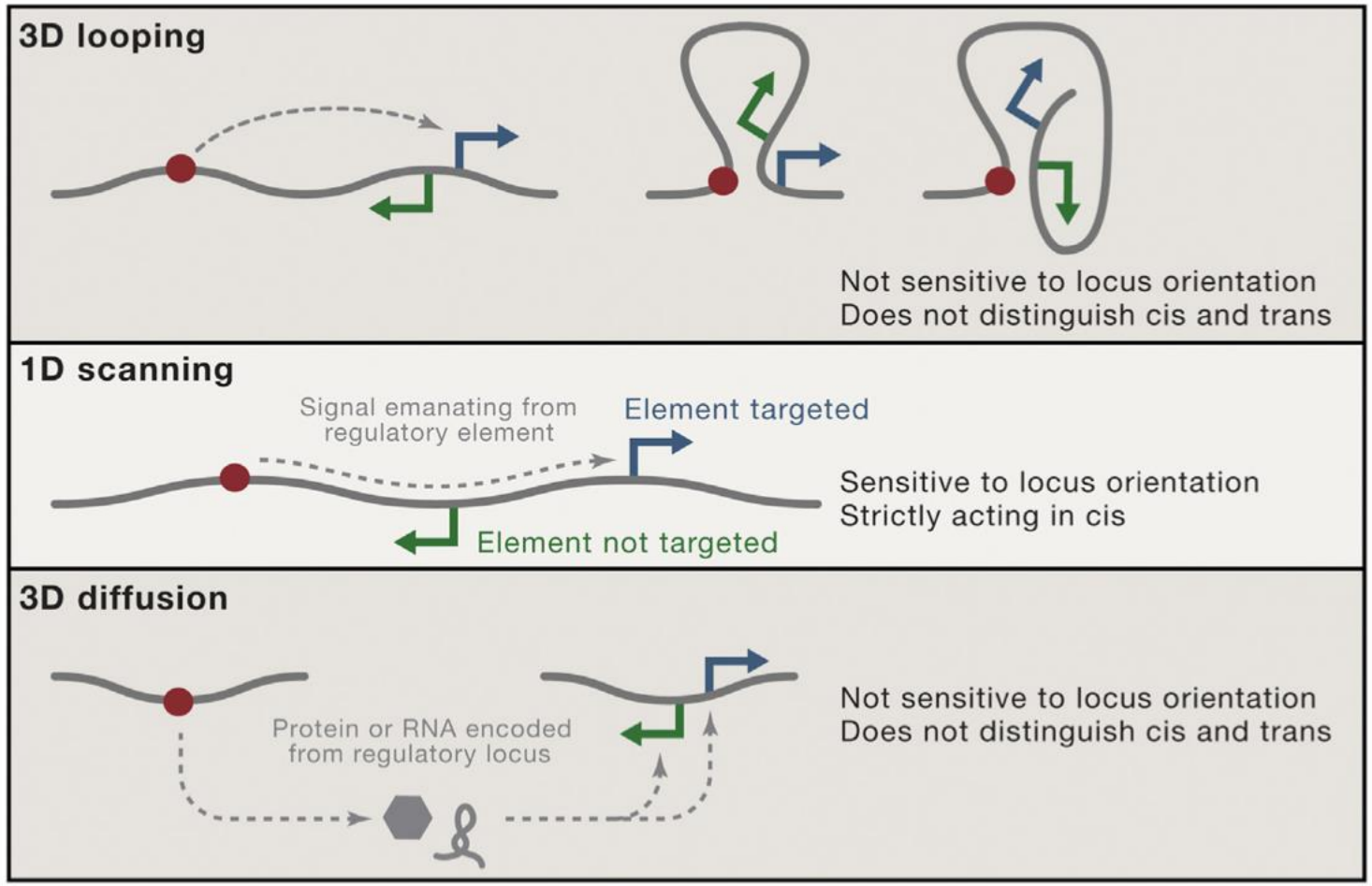


3) Loop extrusion

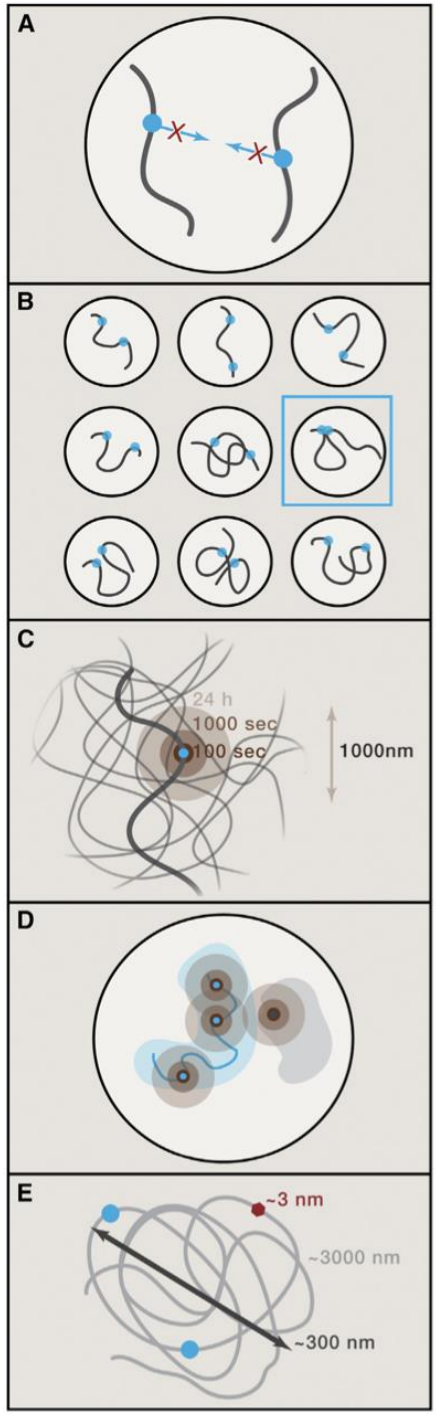


4) Spontaneous interactions

Falta de loops na regulação gênica



Dekker and Mirny, 2016



Alcance limitado

Baixa frequência-variações entre células

Exploração limitada

Interações trans são muito mais variáveis

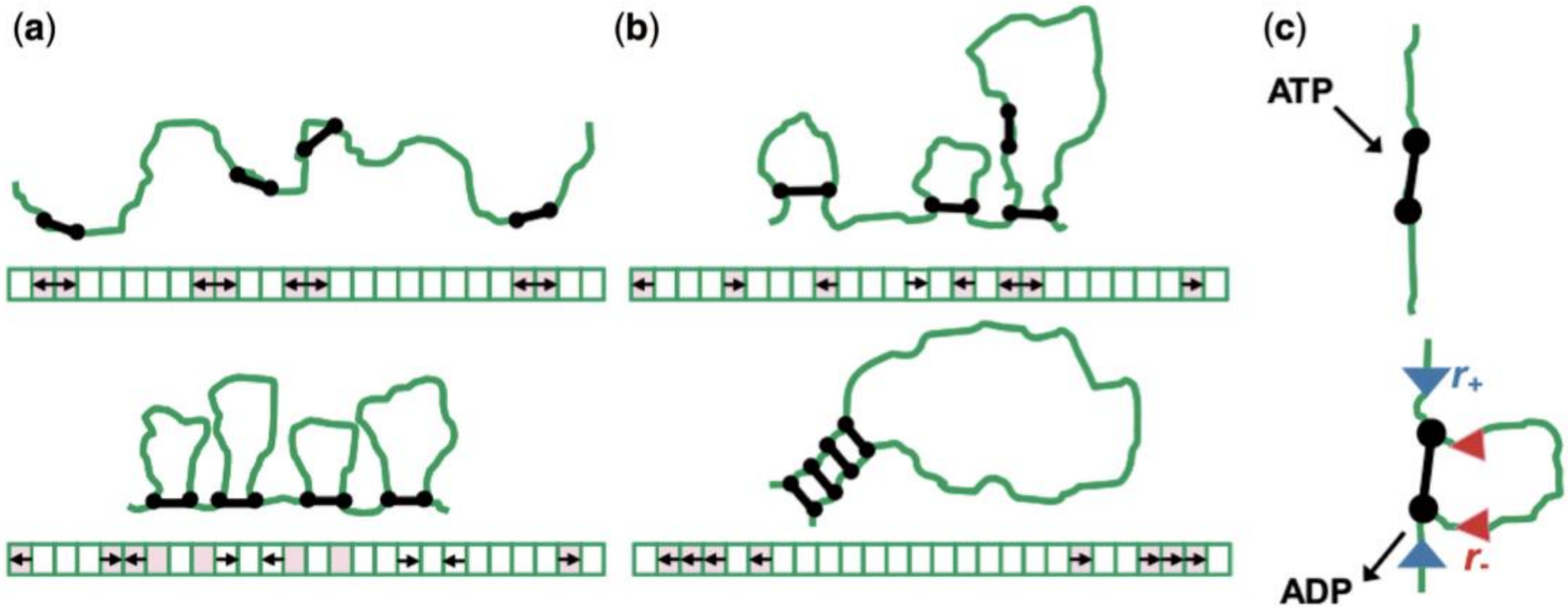
Alcance limitado de proteínas

Self-organization of domain structures by DNA-loop-extruding enzymes

Elnaz Alipour ✉, John F. Marko

Nucleic Acids Research, Volume 40, Issue 22, 1 December 2012, Pages 11202–11212, https://doi.org/10.1093/nar/gks925

Published: 15 October 2012 **Article history** ▾



Structural Maintenance of Chromosomes (SMC) complex proteins

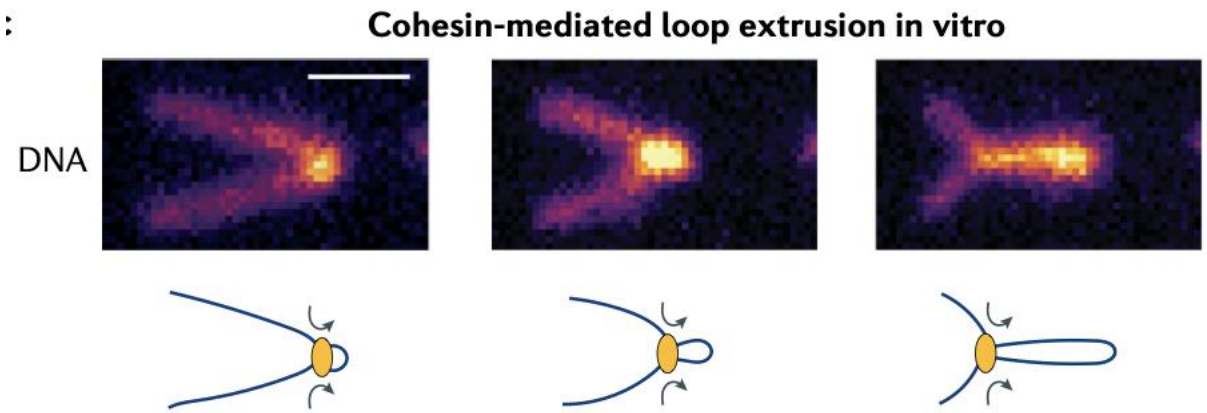
Science

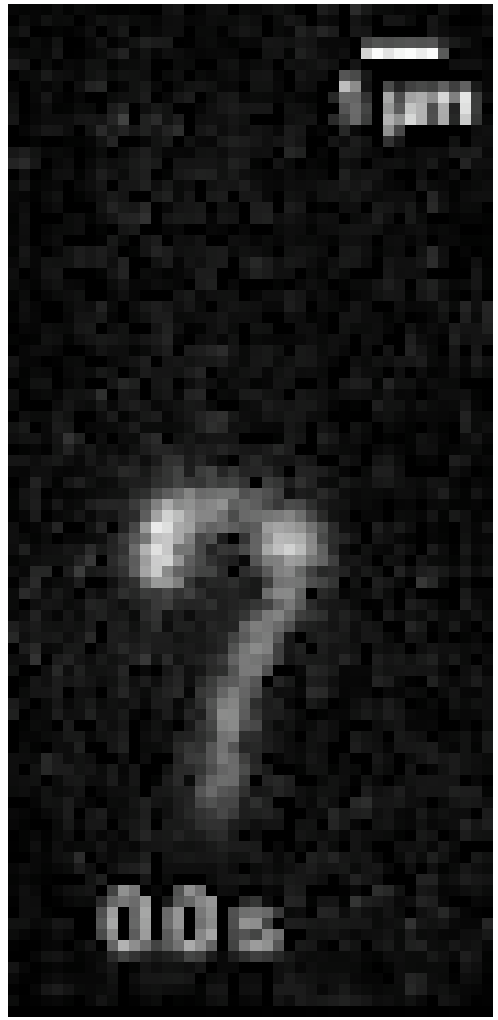
REPORT

Real-time imaging of DNA loop extrusion by condensin

MAHIPAL GANJI, INDRA A. SHALTIEL, SHVETA BISHT, EUGENE KIM, ANA KALICHAVA, CHRISTIAN H. HAERING, AND CEES DEKKER Authors Info & Affiliations

SCIENCE • 22 Feb 2018 • Vol 360, Issue 6384 • pp. 102-105 • DOI: 10.1126/science.aar7831



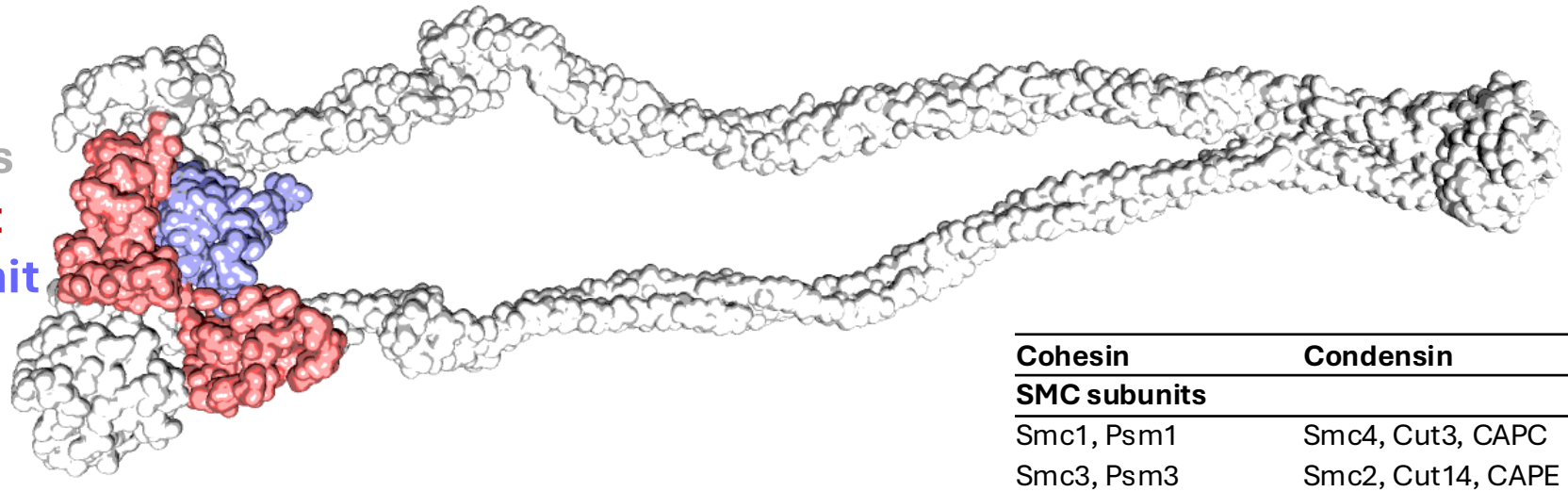


Davidson and Peters, 2021

Structural Maintenance of Chromosomes (SMC) complex proteins

SMC proteins act like chromatin loop extruders.

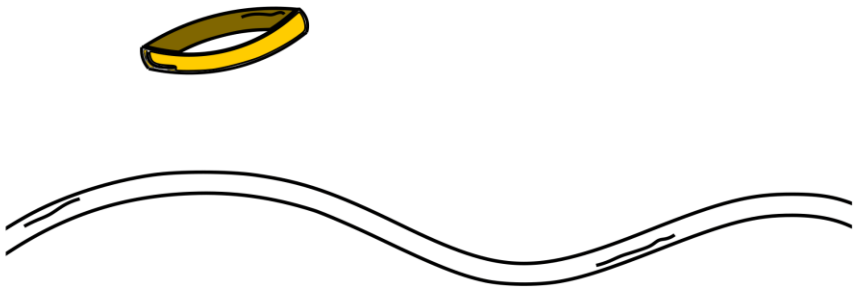
SMC subunits
HEAT subunit
Kleisin subunit



Cohesin	Condensin	Smc5-6 complex
SMC subunits		
Smc1, Psm1	Smc4, Cut3, CAPC	Smc5
Smc3, Psm3	Smc2, Cut14, CAPE	Smc6
Kleisin		
Scc1, Rad21, Rec8	Brn1,Cnd2, CAPH, CAPH2	Nse4, NSMCE4
HEAT subunits		
Scc3, Psc3, SA1, SA2	Ycs4, Cnd1, CAPD2, CAPD3	Nse5, NSMCE5
Pds5, PDS5A, PDS5B	Ycg1, Cnd3, CAPG, CAPG2	Nse6, NSCME6

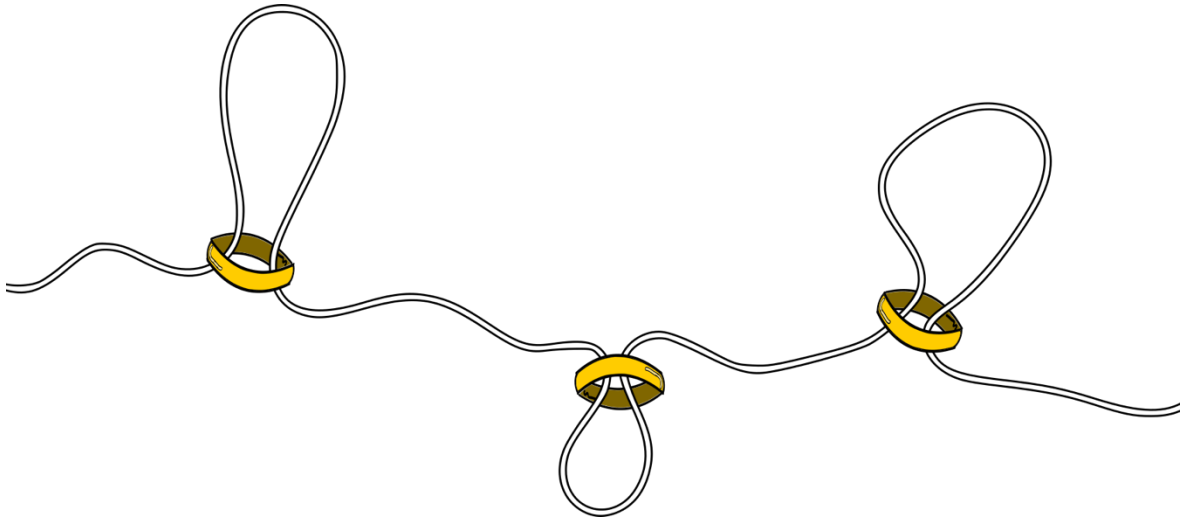
SMC proteins act like chromatin loop extruders

Extrusion: *noun*. The process of forming something by forcing or pushing it out, especially through a small opening.

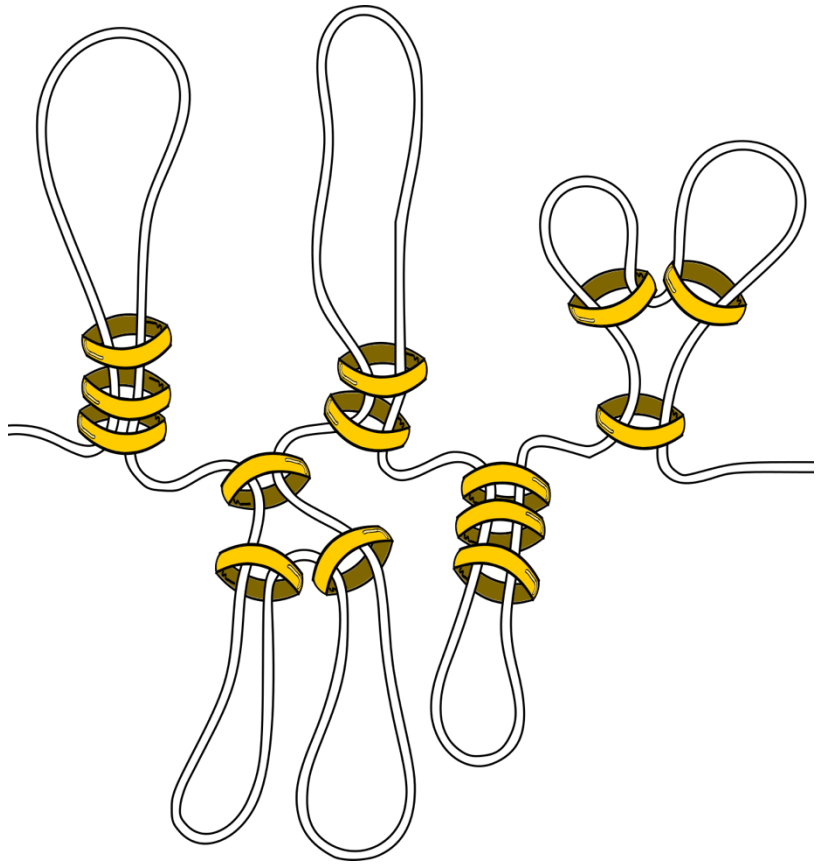


https://youtu.be/mi9WWSy-Nwo

Long distance interactions

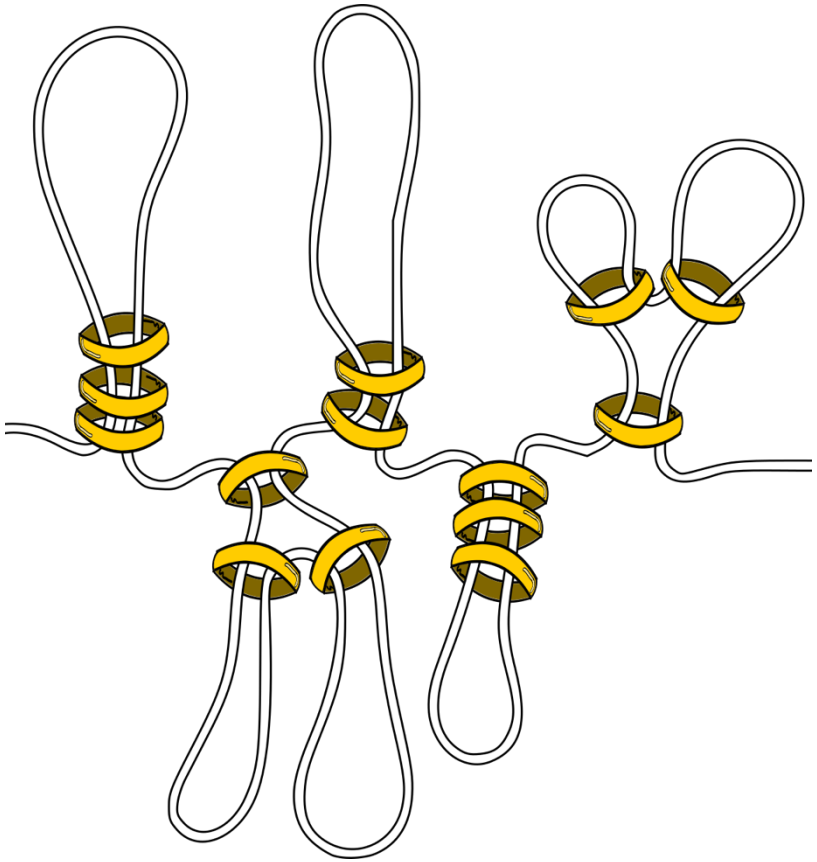


Sparse state



Compact state

How do sister chromatids compact and segregate?



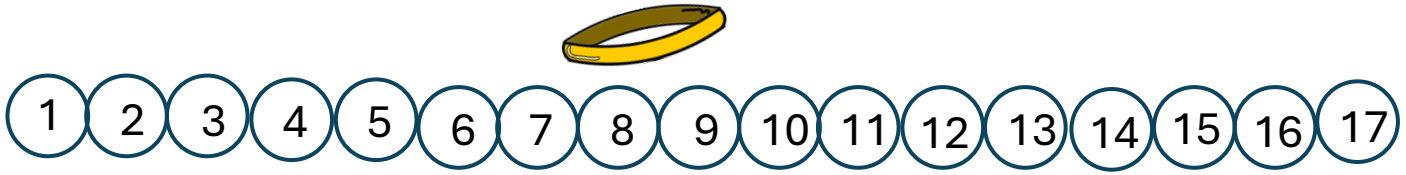
Compact state

Consequences of the compaction by loop extrusion

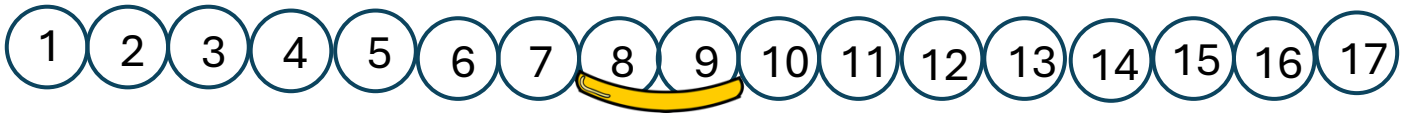
- There are no knots;
- Forms rings, and lacks reptation;
- Subdiffusion;
- Solvent exclusion.

Incorporar extrusão de loops nos modelos

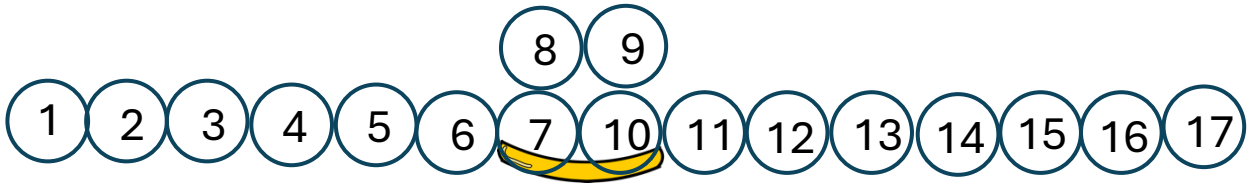
Incorporar extrusão de loops nos modelos



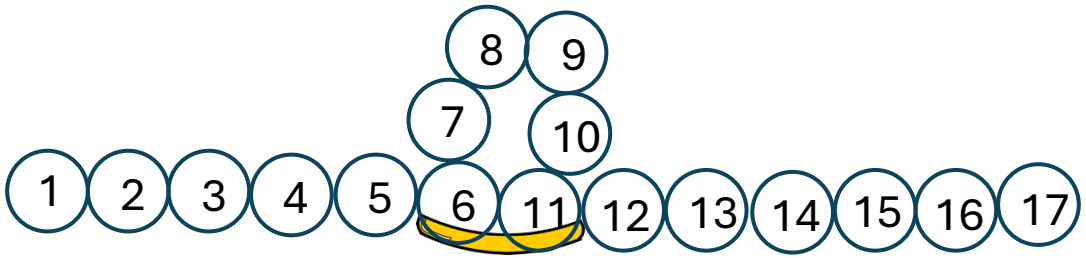
Incorporar extrusão de loops nos modelos



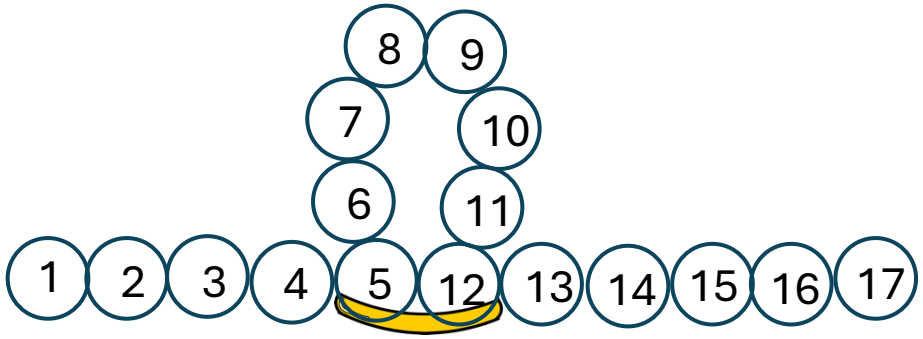
Incorporar extrusão de loops nos modelos



Incorporar extrusão de loops nos modelos

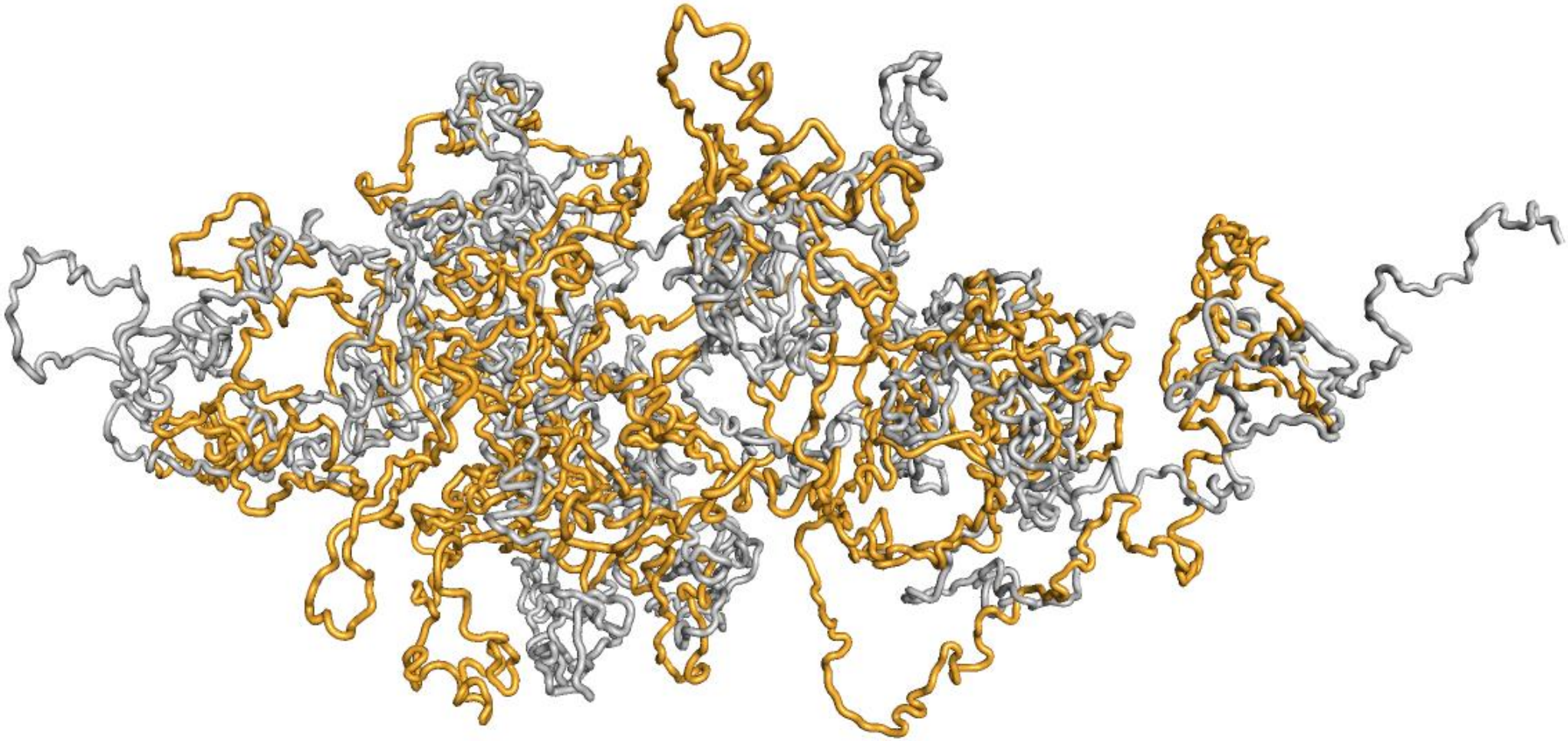


Incorporar extrusão de loops nos modelos

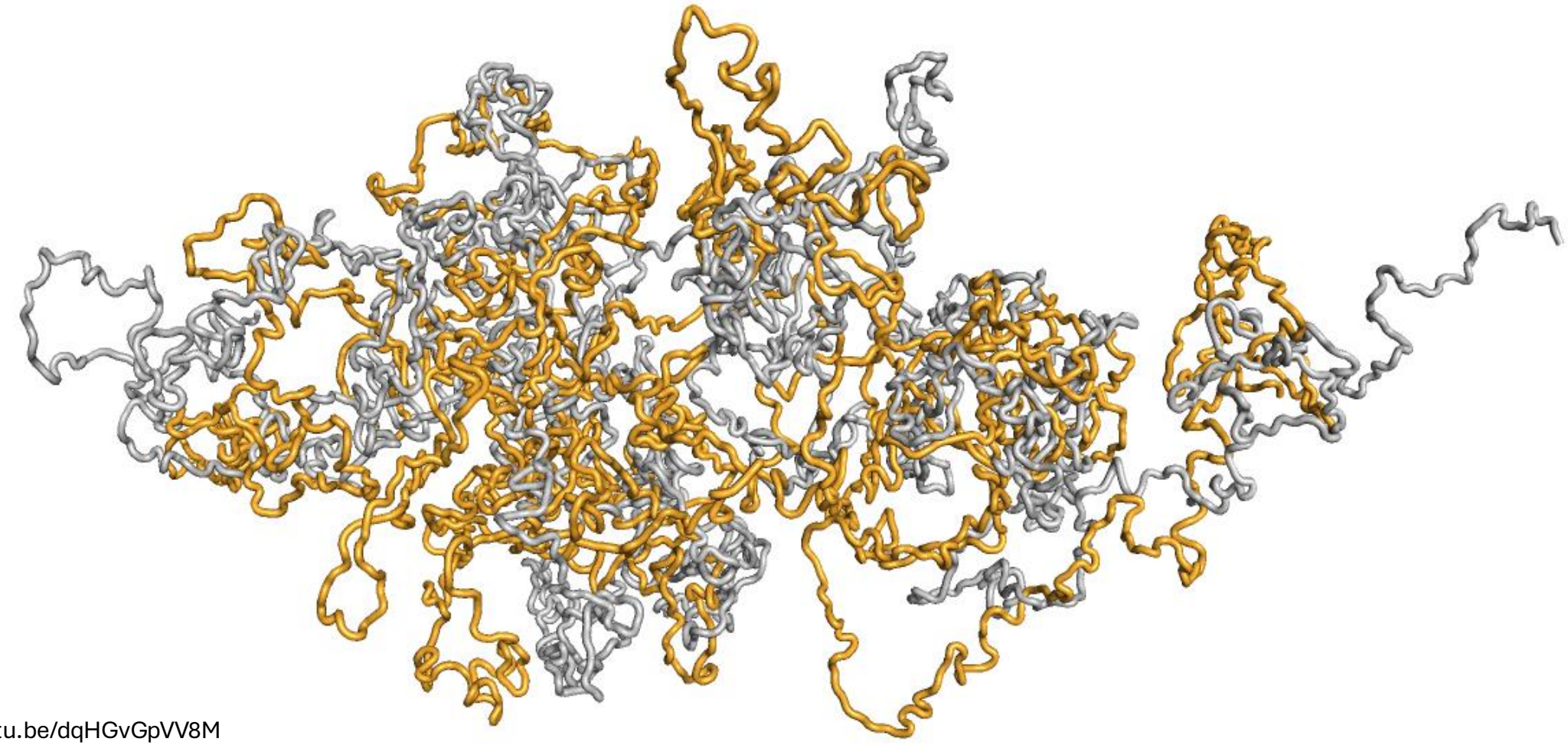


Just a list of pairs that needs to be updated.

Compaction and segregation of sister chromatids

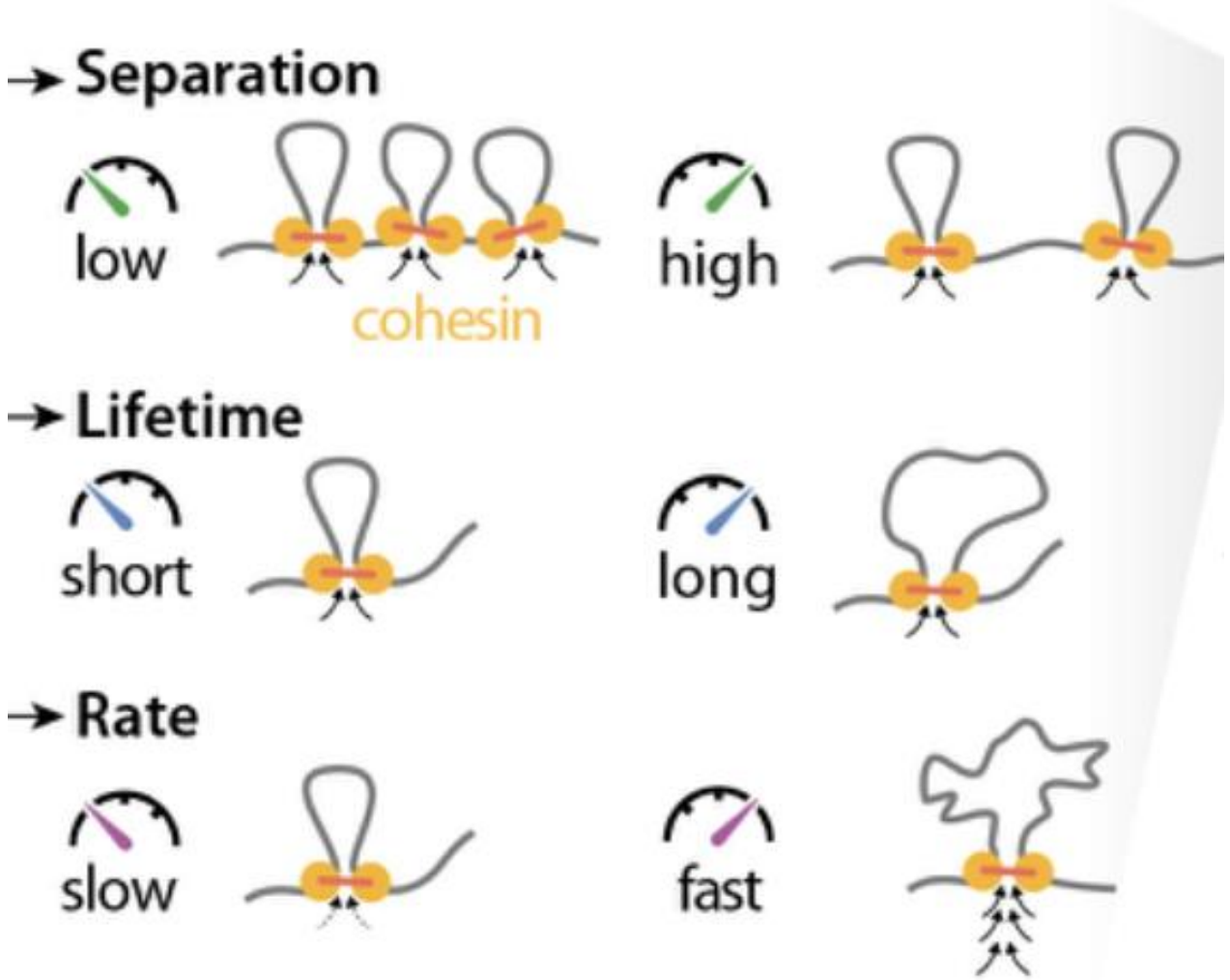


Compaction and segregation of sister chromatids

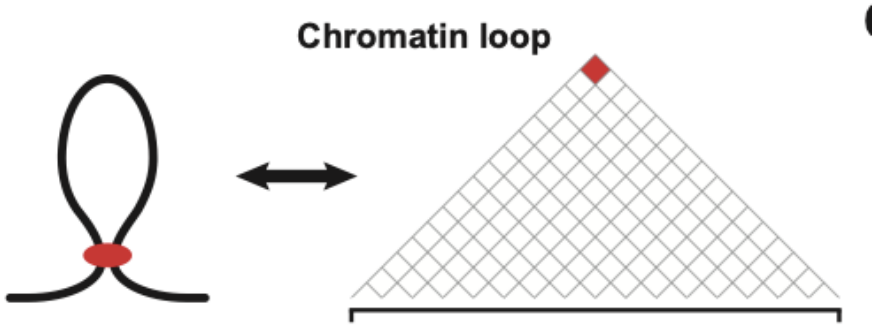


https://youtu.be/dqHGvGpVV8M

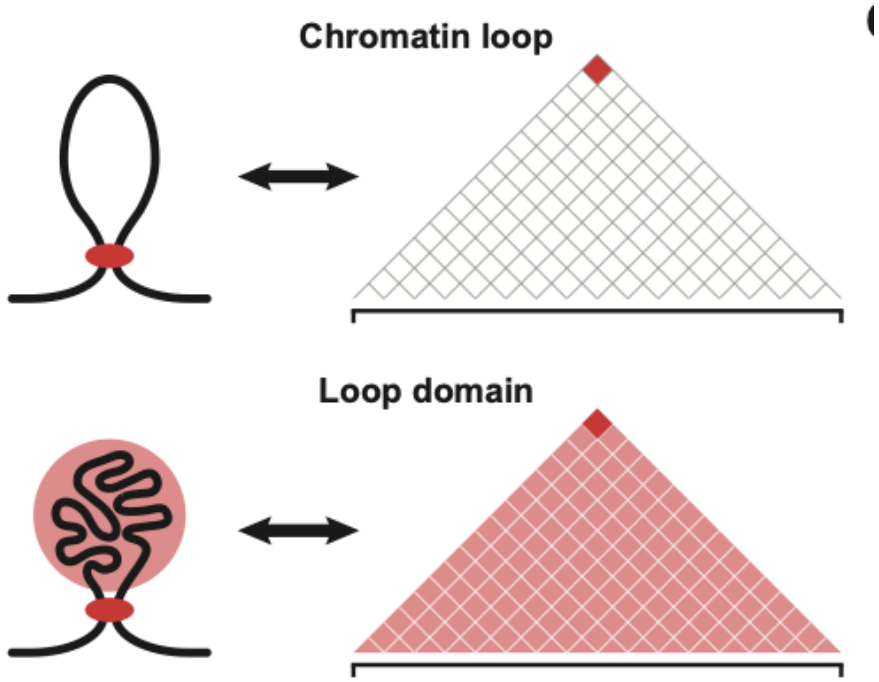
Loops são ajustáveis



Como eles aparecem em mapas de Hi-C



Como eles aparecem em mapas de Hi-C



Como eles aparecem em mapas de Hi-C

